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Neoadjuvant chemotherapy before surgery versus surgery followed by chemotherapy for initial treatment in advanced epithelial ovarian cancer (Review)

Elderfield C, Shawky M, Beazley R, Choudhary C, Coleridge SL, Bryant A, Morrison J

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Neoadjuvant chemotherapy before surgery versus surgery followed by chemotherapy for initial treatment in advanced epithelial ovarian cancer.

Cochrane Database of Systematic Reviews 2026, Issue 7. Art. No.: CD005343.

DOI: [10.1002/14651858.CD005343.pub8](https://doi.org/10.1002/14651858.CD005343.pub8).

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TABLE OF CONTENTS

ABSTRACT	1
PLAIN LANGUAGE SUMMARY	3
SUMMARY OF FINDINGS	5
BACKGROUND	8
OBJECTIVES	9
METHODS	9
Figure 1.	12
Figure 2.	13
RESULTS	14
Figure 3.	15
Figure 4.	19
Figure 5.	20
Figure 6.	21
Figure 7.	22
Figure 8.	25
Figure 9.	26
DISCUSSION	27
AUTHORS' CONCLUSIONS	33
SUPPLEMENTARY MATERIALS	34
ADDITIONAL INFORMATION	34
REFERENCES	37
ADDITIONAL TABLES	44

[Intervention Review]

Neoadjuvant chemotherapy before surgery versus surgery followed by chemotherapy for initial treatment in advanced epithelial ovarian cancer

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Editorial group: Cochrane Central Editorial Service.

Publication status and date: New search for studies and content updated (no change to conclusions), published in Issue 7, 2026.

Citation: Elderfield C, Shawky M, Beazley R, Choudhary C, Coleridge SL, Bryant A, Morrison J. Neoadjuvant chemotherapy before surgery versus surgery followed by chemotherapy for initial treatment in advanced epithelial ovarian cancer. *Cochrane Database of Systematic Reviews* 2026, Issue 7. Art. No.: CD005343. DOI: [10.1002/14651858.CD005343.pub8](https://doi.org/10.1002/14651858.CD005343.pub8).

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ABSTRACT

Rationale

Epithelial ovarian cancer (EOC) presents at an advanced stage in the majority of women. These women require a combination of surgery and chemotherapy for optimal treatment. Conventional treatment has been to perform surgery first and then give chemotherapy. However, there may be advantages to using chemotherapy before surgery. This is an update of a review first published in 2007 and last updated in 2025.

Objectives

To assess the advantages and disadvantages of treating women with advanced EOC with chemotherapy before cytoreductive surgery (neoadjuvant chemotherapy (NACT)) compared with conventional treatment where chemotherapy follows cytoreductive surgery (primary cytoreductive surgery (PCRS)).

Search methods

We searched CENTRAL, MEDLINE, Embase, ClinicalTrials.gov, and the World Health Organization International Clinical Trials Registry Platform on 9 October 2025. We also checked the reference lists of relevant papers for further studies. We contacted the principal investigators of relevant studies for further information.

Eligibility criteria

Randomised controlled trials (RCTs) of women with advanced EOC (International Federation of Gynecology and Obstetrics (FIGO) stage III/IV) who were randomly allocated to treatment groups that compared platinum-based chemotherapy before cytoreductive surgery with platinum-based chemotherapy following cytoreductive surgery.

Outcomes

We extracted data on overall survival and progression-free survival, adverse events, surgically related mortality and morbidity, and quality-of-life outcomes.

Risk of bias

We used the Cochrane Risk of Bias 1 tool (RoB 1) to assess risk of bias in RCTs.

Synthesis methods

We conducted meta-analyses using random-effects models (due to heterogeneity between studies) to calculate hazard ratios (HR), risk ratios (RR), mean differences (MD), and 95% confidence intervals (CI) for all outcomes. We assessed the certainty of evidence according to the GRADE approach.

Included studies

A total of seven RCTs of varying quality and size met the inclusion criteria, with two new completed studies in this update. The studies assessed a total of 2650 women with stage III/IV ovarian cancer randomised to NACT followed by interval cytoreductive surgery (ICRS) or PCRS followed by chemotherapy. We included data from five studies in the meta-analyses (2380 participants).

Synthesis of results

Survival

We found little or no difference between groups in overall survival (HR 1.00, 95% CI 0.91 to 1.10; $I^2 = 2\%$; 5 studies, 2380 participants; high-certainty evidence) and likely little or no difference between groups in progression-free survival (HR 1.03, 95% CI 0.92 to 1.15; $I^2 = 34\%$; 5 studies, 2380 participants; moderate-certainty evidence).

Adverse events

Adverse events, surgical morbidity, and quality-of-life outcomes were variably and incompletely reported across studies. NACT reduces postoperative mortality (0.5% in the NACT group versus 2.6% in the PCRS group) (RR 0.25, 95% CI 0.10 to 0.63; $I^2 = 0\%$; 5 studies, 2201 participants; high-certainty evidence). There are probably clinically meaningful differences in favour of NACT compared to PCRS in overall surgically related adverse effects (grade 3+ (G3+)) (6.4% in the NACT group versus 29.4% in the PCRS group) (RR 0.34, 95% CI 0.14 to 0.82; $I^2 = 82\%$; 3 studies, 1094 participants; moderate-certainty evidence).

Organ resection

NACT probably results in a large reduction in the need for stoma formation (6.9% in the NACT group versus 20.1% in the PCRS group) (RR 0.34, 95% CI 0.20 to 0.57; $I^2 = 54\%$; 3 studies, 1291 participants; moderate-certainty evidence) and probably reduces the risk of needing bowel resection at the time of surgery (18.9% in the NACT group versus 38.6% in the PCRS group) (RR 0.49, 95% CI 0.35 to 0.69; $I^2 = 79\%$; 5 studies, 2237 participants; moderate-certainty evidence).

Quality of life

Global quality of life on the EORTC QLQ-C30 produced imprecise results in three studies, with high levels of heterogeneity (quality of life at 6 months: MD 6.62, 95% CI -2.89 to 16.13; $I^2 = 92\%$; 3 studies, 559 participants; very low-certainty evidence) and any differences might not be clinically meaningful.

Authors' conclusions

The available high- to moderate-certainty evidence shows there is likely little or no difference in primary survival outcomes between PCRS and NACT for those with advanced EOC who are suitable for either treatment option. NACT reduces the risk of postoperative mortality, and probably reduces the risk of serious adverse events around the time of surgery, and the need for stoma formation. These data are consistent across ~20 years of studies, through significant changes in surgical radicality, and should inform women and clinicians (involving specialist gynaecological multidisciplinary teams) to allow treatment to be tailored to the individual patient, taking into account surgical resectability, age, histology, stage, and performance status. Further data from studies unpublished in peer-reviewed journals and ongoing studies are awaited, but are unlikely to significantly change the results of this review.

Funding

This Cochrane review update had no dedicated funding.

Registration

Protocol (2005): DOI: 10.1002/14651858.CD005343

Original review (2007): DOI: 10.1002/14651858.CD005343.pub2

Review update (2012): DOI: 10.1002/14651858.CD005343.pub3

Review update (2019): DOI: 10.1002/14651858.CD005343.pub4

Review update (2021): DOI: 10.1002/14651858.CD005343.pub5

Review updated (2021a): DOI: 10.1002/14651858.CD005343.pub6

Review updated (2025): DOI: 10.1002/14651858.CD005343.pub7

PLAIN LANGUAGE SUMMARY

Does giving chemotherapy before surgery improve survival and quality of life in women with advanced epithelial ovarian cancer?

Key messages

- There is little difference in how long women with advanced (stage IIIC/IV) epithelial ovarian cancer (EOC) survive, whether they have chemotherapy or surgery first. There is probably little difference in how long it takes for EOC to return after treatment.
- Giving chemotherapy prior to surgery probably reduces serious risks of surgery; probably halves the risk of needing bowel removal during surgery; and probably results in a large reduction in needing a stoma (where the bowel is diverted through the abdominal wall into a bag to collect the contents).
- Giving chemotherapy prior to surgery is an alternative to surgery followed by chemotherapy in women with advanced EOC. Decisions about which treatment to have first depend on patient preference, how well the woman is at time of diagnosis, the risks of surgery, and the amount and spread of disease.

What is epithelial ovarian cancer, and how is it treated?

Ovarian cancer is the eighth most common cancer worldwide in women. Around 90% of ovarian cancers are epithelial ovarian cancer (EOC), arising from the surface of the ovary or lining of fallopian tubes. Most women with EOC are diagnosed when their cancer is at a late stage, and their disease has spread throughout the abdominal cavity (stage IIIC/IV). Although survival rates have improved over the last 30 years, only four in every 10 women with EOC are alive 10 years after diagnosis.

Treatment for ovarian cancer involves a combination of surgery and chemotherapy. Surgery aims to remove as much visible cancer as possible. However, surgery alone is unlikely to cure EOC, and most women will also need chemotherapy. Chemotherapy uses platinum-based medications to treat cells that cannot be removed by surgery or cannot be seen.

Traditionally, chemotherapy is given after surgery. However, chemotherapy can be given before surgery and then followed by the remaining cycles of chemotherapy, as usual.

Why is this important?

Women with advanced EOC may be too unwell at diagnosis to have surgery, or may require extensive surgery to remove all visible disease. Giving chemotherapy before surgery may shrink the cancer prior to surgery, so that women become well enough to undergo surgery or need less extensive surgery.

What did we want to find out?

We wanted to find out if giving chemotherapy before surgery was better than doing surgery first and then giving chemotherapy. We were interested in how many women with EOC:

- were alive 48 months after diagnosis;
- had no EOC 24 months after treatment;
- died following surgery;
- had unwanted effects due to surgery;
- needed to have their bowel diverted through the abdominal wall into a bag to collect bowel contents (a 'stoma'), or to have part of the bowel removed (a 'bowel resection'); and
- saw an improvement in their quality of life.

What did we do?

We searched for studies that investigated chemotherapy followed by surgery compared with surgery followed by chemotherapy in women diagnosed with advanced EOC. We compared and summarised the results of the studies and rated our confidence in the evidence based on factors such as study methods and sizes.

What did we find?

We found seven studies with 2650 women with stage III/IV EOC that compared the effects of giving an initial chemotherapy treatment followed by surgery then chemotherapy as normal, with surgery followed by chemotherapy. Studies took place worldwide, in 20, mainly high-income countries.

Giving chemotherapy before surgery:

- makes little or no difference to the number of women still alive after 48 months (5 studies, 2380 women);
- probably makes little or no difference to the number of women who remained disease-free after 24 months (5 studies, 2380 women);
- reduces deaths due to surgery (5 studies, 2201 women);
- probably reduces the risk of severe unwanted effects of surgery (3 studies, 1094 women);
- probably results in a large reduction in the risk of needing a stoma (3 studies, 1291 women), or a bowel resection (5 studies, 2237 women);
- may slightly improve quality of life 6 months after treatment, but the results are uncertain (3 studies, 559 women).

What are the limitations of the evidence?

We are confident that chemotherapy before surgery makes no difference to overall survival and reduces deaths due to surgery. We are less confident in the rest of the evidence for several reasons. It was clear which treatment women received, which might change the results. Although the studies only enrolled women with advanced EOC (stage III/IV), and many had extensive disease, later studies only enrolled women when the surgeons thought they could remove all of the visible disease with surgery. We could only analyse results from five of the seven studies. We will add results from further studies when they are available.

How up-to-date is the evidence?

The evidence is current to 9 October 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Neoadjuvant chemotherapy (NACT) followed by interval cytoreductive surgery (ICRS) compared to primary cytoreductive surgery (PCRS) followed by chemotherapy for women with advanced epithelial ovarian cancer (EOC)

Neoadjuvant chemotherapy (NACT) followed by interval cytoreductive surgery (ICRS) compared to primary cytoreductive surgery (PCRS) followed by chemotherapy for women with advanced epithelial ovarian cancer (EOC)

Patient or population: women with advanced epithelial ovarian cancer (EOC)

Setting: tertiary centres treating women with EOC

Intervention: neoadjuvant chemotherapy (NACT) followed by interval cytoreductive surgery (ICRS)

Comparison: primary cytoreductive surgery (PCRS) followed by chemotherapy

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with primary cytoreductive surgery (PCRS) followed by chemotherapy	Risk with neoadjuvant chemotherapy (NACT) followed by interval cytoreductive surgery (ICRS)				
Overall survival assessed with: number alive at 48 months' follow-up follow-up: range 4.4 years to 6 years	Moderate 286 per 1000	286 per 1000 (252 to 320)	HR 1.00 (0.91 to 1.10) [overall survival]	2380 (5 RCTs)	⊕⊕⊕⊕ High ^a	NACT results in little to no difference in overall survival. Absolute risk of survival at 4 years demonstrated for absolute effects using formula of corresponding intervention risk per 1000 = 1000-(exp[ln(1-proportion of patients without event) x HR]) x 1000 (Tierney 2007). Control survival estimates (moderate) taken from SCORPION study. ^{1,2,a,b}
Progression-free survival assessed with: number without disease progression at 24 months' follow-up follow-up: range 4.4 years to 6 years	Moderate 310 per 1000	299 per 1000 (260 to 340)	HR 1.03 (0.92 to 1.15) [survival without disease progression]	2380 (5 RCTs)	⊕⊕⊕⊖ Moderate ^b	NACT probably results in little to no difference in progression-free survival. Absolute risk of survival without disease progression at 2 years demonstrated for absolute effects using formula of corresponding intervention risk per 1000 = 1000-(exp[ln(1-proportion of patients without event) x HR]) x 1000 (Tierney 2007). Control progression-free survival (PFS) estimates (moderate) taken from SCORPION study. ^{1,2}

Postoperative mortality	26 per 1000	6 per 1000 (3 to 16)	RR 0.25 (0.10 to 0.63)	2201 (5 RCTs)	⊕⊕⊕⊕ High ^a	NACT followed by ICRS results in large reduction in postoperative mortality ^{a,b}
Surgically-related adverse effects (G3+); overall postoperative adverse events (G3+)	222 per 1000	76 per 1000 (31 to 182)	RR 0.34 (0.14 to 0.82)	1094 (3 RCTs)	⊕⊕⊕⊖ Moderate ^b	NACT followed by ICRS probably results in a large reduction in overall postoperative adverse events (G3+). ^c
Surgical resection: stoma formation follow-up: range 0 days to 0 days	201 per 1000	69 per 1000 (40 to 115)	RR 0.34 (0.20 to 0.57)	1291 (3 RCTs)	⊕⊕⊕⊖ Moderate ^b	NACT followed by ICRS probably results in a large reduction in need for stoma formation. ^c
Surgical resection: bowel resection follow-up: range 0 days to 30	386 per 1000	189 per 1000 (135 to 266)	RR 0.49 (0.35 to 0.69)	2237 (5 RCTs)	⊕⊕⊕⊖ Moderate ^b	NACT followed by ICRS probably results in a large reduction in need for bowel resection. ^c
EORTC QLQ-C30 quality of life at 6 months: functional status - global health	The mean EORTC QLQ-C30 quality of life at 6 months: functional status - global health was 62.17 EORTC QLQ-C30 global health score	MD 6.62 EORTC QLQ-C30 global health score higher (2.89 lower to 16.13 higher)	-	559 (3 RCTs)	⊕⊖⊖⊖ Very low ^{b,d,e}	The evidence is very uncertain about the effect of NACT on EORTC QLQ-C30 QoL at 6 months: functional status - global health.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **HR:** hazard ratio; **MD:** mean difference; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_464099510081044711.

^a Lack of blinding unlikely to have influenced outcome therefore not downgraded for risk of bias due to lack of blinding.

- ^b Downgraded by one level for risk of bias in outcome due to lack of blinding.
- ^c Rate of adverse events differs between studies due to change in degree of surgical radicality over time, but direction and relative degree consistent, so not downgraded for inconsistency.
- ^d Downgraded by one level due to concerns about heterogeneity.
- ^e Downgraded by one level due to imprecision (95% CI that means outcomes could be either better or worse with NACT/ICRS and variability between studies).
- ¹ Fagotti, A., Ferrandina, M. G., Vizzielli, G., Pasciuto, T., Fanfani, F., Gallotta, V., Margariti, P. A., Chiantera, V., Costantini, B., Gueli Alletti, S., Cosentino, F., Scambia, G.. Randomized trial of primary debulking surgery versus neoadjuvant chemotherapy for advanced epithelial ovarian cancer (SCORPION-NCT01461850) .Int J Gynecol Cancer ; 2020.
- ² Tierney JF, Stewart LA, Gherzi D, Burdett S, Sydes MR.. Practical methods for incorporating summary time-to-event data into meta-analysis. .Trials; 2007.

BACKGROUND

Description of the condition

Ovarian cancer is now the eighth most common cancer in women, affecting 324,398 women globally in 2022 [1]. Despite survival rates improving by 5% to 20% over the past 30 years, two in five women in Europe and North America with ovarian cancer are alive 10 years after diagnosis [2, 3, 4, 5, 6].

Epithelial ovarian cancer (EOC) is the most common form of ovarian cancer (~90% of ovarian cancers), arising from the surface of the ovary or lining of fallopian tubes (as reviewed by Kyo and colleagues [7]). Most women with EOC present at a late stage, when the cancer has already spread throughout the abdominal cavity and possibly further afield, to the lining of the lung cavity (pleural) or lymph nodes outside the abdomen [8]. Symptoms are often vague, with women presenting to primary care practitioners for several months before diagnosis [9]. There are no effective screening programmes [10]. Abnormal cells can spread throughout the abdominal cavity, even when the primary tumour is microscopic, implant on other surfaces, and grow over time until they cause symptoms.

Description of the intervention and how it might work

In early-stage disease (International Federation of Gynecology and Obstetrics (FIGO) stage I/IIa; Table 1; Table 2), surgery will cure most women, although a proportion of women still benefit from adjuvant chemotherapy [11]. In advanced cancer, even 'maximal effort' surgery cannot remove all microscopic disease, and so survival is dependent upon chemo-sensitivity. Unfortunately, around 75% of women present when EOC has spread outside the pelvis (FIGO stage III/IV), when surgery alone cannot be curative, and the optimal timing of surgery is less clear. The majority of those with EOC have high-grade serous ovarian cancer (~7 in 10). Other histological subtypes are rarer (e.g. endometrioid, clear cell, mucinous and low-grade serous) and have relative chemo-insensitivity.

The standard treatment for women with advanced ovarian cancer (FIGO stage III/IV) is a staging laparotomy with primary cytoreduction surgery (PCRS) followed by platinum-based chemotherapy. The extent of tumour cytoreduction is considered the most important prognostic factor. Griffiths 1975 was the first to report a relationship between the size of residual disease and survival [12]. Meta-analyses of non-randomised studies (NRS) have since concurred that survival correlates positively with the extent of tumour cytoreduction achieved [13, 14, 15]. However, the extent of cytoreduction achievable may be directly related to tumour biology, which would strongly bias results from NRS. Tumours that have also spread to the para-aortic, mediastinal, or scalene (supraclavicular) lymph nodes may be less likely to be optimally cytoreduced by surgery [16, 17]. The ability to achieve successful cytoreduction may thus in part reflect tumour biology. One exploratory analysis of three prospectively randomised trials in advanced ovarian cancer suggested that surgical cytoreduction can partially overcome these biological factors [18]. Other independent prognostic factors for overall survival were shown to be age, performance status, grade, FIGO stage, and histology [18]. Interestingly, a recent study demonstrated that routinely removing non-bulky lymph nodes in EOC does not improve survival [19].

The definition of what constitutes 'optimal' or 'maximal' cytoreduction has changed since the 1980s. Originally considered

to be no residual tumour deposit larger than 2 cm in diameter, and more recently as a residual tumour of 1 cm or smaller, the current aim is to leave no macroscopic residual disease (NMRD), that is, no disease left visible to the naked eye; so-called 'complete' or 'R0' surgery [20]. However, this is not conventional R0 resection; in other cancers, an 'R0 resection' indicates that the tumour has been removed with proven microscopically normal margins. In advanced ovarian cancer, due to the pattern of spread via the intra-abdominal cavity, in disease that is advanced at presentation, microscopic disease is likely to remain even after an NMRD is achieved, hence, we do not use the terms 'complete' and 'R0' in this review.

In the past, some investigators had not shown a benefit to maximal cytoreduction in women with high-volume, advanced disease [21; 22]. However, this may have been because some of these women were very unwell prior to surgery and not fit enough at that stage to withstand a major operation. In their 1998 study, Vergote and colleagues therefore introduced a policy of treating women with primary chemotherapy (neoadjuvant chemotherapy (NACT)) or PCRS, depending on the extent of the disease and performance status. Following the change in patient management, they reported an overall improvement in survival, despite a reduction in PCRS rates from 82% to 57%.

The role of so-called 'maximal effort' surgery in ovarian cancer, with extensive surgery, often involving the upper abdomen, is a separate question, and this review did not seek to question the value or extent of surgery, but rather its timing with respect to its combination with chemotherapy. However, a non-randomised study demonstrated the importance of the combination of surgery and chemotherapy, with reduced survival in those who had chemotherapy alone and did not go on to have interval cytoreductive surgery (ICRS) [23]. This is supported by findings from a cohort study from Sweden, which demonstrated no improvement in survival with system-wide introduction of ultra-radical surgery for ovarian cancer, associated with a reduction in those undergoing surgery by around 10% [24]. Studies that do not use whole-population cohorts are at critical risk of bias and may overestimate the benefits of PCRS (e.g. Mueller and colleagues [25]). NACT involves giving chemotherapy before attempting cytoreductive surgery (CRS) for advanced ovarian cancer and is a rationale used in other tumour types. It has evolved from the practice of ICRS, a secondary attempt at tumour cytoreduction performed after a suboptimal attempt at PCRS and adjuvant chemotherapy. In a Cochrane review, additional ICRS performed by gynaecological oncologists secondary to PCRS and adjuvant chemotherapy was found to offer no additional survival benefit compared with standard treatment of advanced ovarian cancer [26]. However, ICRS may improve survival of women in whom primary surgery was not performed with cytoreductive intent by a gynaecological oncologist and in those who have had residual disease larger than 2 cm at primary surgery.

Bristow and colleagues reviewed 26 NRS comparing NACT with PCRS and concluded that, while NACT might be a viable option for those unsuitable for an attempt at primary cytoreduction due to significant comorbidities, current poor performance status, or impossibility of surgery, survival outcomes with NACT may be inferior to PCRS [27]. However, this was based on highly selected data, at critical risk of bias, as women with worse disease were more likely to have received NACT/ICRS rather than PCRS. Platinum-based NACT may thus be an alternative to PCRS, particularly

where complete cytoreduction at PCRS is considered unlikely [28]. Tumour resectability depends on the patient's age, disease burden, comorbidities, location of metastatic sites, performance status and stage [29], as well as the skill and philosophy of the surgical team [30, 31, 32]. Retrospective data suggest that the optimal time for ICPS may be after three cycles of chemotherapy, followed by a further three cycles, and that delaying to four cycles might worsen overall survival [33]. However, these data are based on retrospective analysis of NRS data and are therefore at critical risk of bias, as women who are less well are more likely to have delayed surgery. On multivariate analysis, only the ECOG performance status correlated with overall survival (hazard ratio 1.76, 95% confidence interval 1.2 to 2.49; $P = 0.001$).

The goal of surgery, whether ICPS or PCRS, should be complete resection of all disease [34]. A review of 21 NRS found that, compared with PCRS, NACT improved the rate of optimal cytoreduction [35]. However, this did not seem to influence survival.

There are several reasons why NACT may be preferable to PCRS:

- NACT may decrease the size and extent of the tumour such that complete resection is more feasible;
- NACT may improve patient performance status;
- PCRS necessitates hospital admission, whereas chemotherapy can be administered in an outpatient setting and started immediately;
- PCRS delays starting chemotherapy, as there is the potential for chemotherapy to interfere with wound healing;
- if surgery is not curative, residual tumour cells may multiply while the individual awaits recovery from surgery.

Concerns about using NACT include the following:

- NACT delays the removal of the tumour and may therefore compromise patient survival;
- chemotherapy induces fibrosis, which may make CRS more difficult;
- NACT may effectively shrink cancer deposits but leave microscopic disease that is then not surgically removed, whereas the whole deposit might have been removed had it been visible;
- if too many cycles of NACT are given pre-surgery, there is a concern regarding the possibility of chemo-resistance post-surgery. One meta-analysis found a negative association between overall survival and the number of NACT cycles given [36];
- PCRS reduces the tumour bulk and number of cancer cells, thereby reducing the chance of developing chemo-resistance.

Why it is important to do this review

There has been considerable controversy surrounding the use of NACT in advanced ovarian cancer [29, 37, 38]. One early overview stated, "NACT is expected to become standard treatment for unselected women with advanced ovarian cancer when favourable results are confirmed by Phase III studies and several problems are resolved" [39]. However, surveys among members of the US Society of Gynecologic Oncology [40], and the European Society of Gynaecologic Oncology [32], suggest a large discrepancy in the acceptance and use of NACT as a treatment option for advanced

ovarian cancer. In the UK, there is also significant variation in practice in terms of the proportion of patients receiving any treatment, surgery, and scheduling of surgery with chemotherapy (i.e. NACT versus PCRS) in a national feasibility audit [41], and the more recent National Ovarian Cancer Audit [42]. Many investigators agree that NACT has a place, at the very least, in women with lesions that cannot be optimally resected, or in those too unwell to undergo major surgery at diagnosis [27, 28, 29, 30]. Interestingly, a difference-in-difference study demonstrated that centres in the USA that changed practice and significantly increased the use of NACT following the publication of EORTC-55971 [43], improved survival for women with EOC to a greater extent than those that did not change practice [44]. Since RCTs are the 'gold standard' of evidence-based medical research, we hope that a review of randomised evidence may clarify the benefits and risks of using NACT for women with advanced ovarian cancer, compared with the standard treatment of PCRS.

This is an update of a Cochrane review first published in 2007 and last updated in 2025 [45, 46, 47, 48, 49, 50].

OBJECTIVES

To assess the advantages and disadvantages of treating women with advanced epithelial ovarian cancer (EOC) with chemotherapy before cytoreductive surgery (neoadjuvant chemotherapy (NACT)) compared with conventional treatment where chemotherapy follows cytoreductive surgery (primary cytoreductive surgery (PCRS)).

METHODS

We followed the Methodological Expectations of Cochrane Intervention Reviews (MECIR) requirements when conducting the review [51], and PRISMA 2020 for the reporting [52], including the addition of the GRADE approach and a summary of findings table [53].

Differences between protocol and review

The original review protocol was published in 2005, and the review has evolved over time to incorporate previous reviewer feedback, and newer methods of assessing and reporting the risk of bias in included studies.

In response to requests by previous reviewers, we included subgroup analyses on survival by age, stage, and residual disease in previous updates, and these are included in this review update as prespecified items; we also added perioperative death, bowel resection, and stoma formation to the outcome measures and included these in [Summary of findings 1](#), as these are important outcomes for women. In a previous update, we included postoperative death, and included it in the current review as a prespecified outcome in [Summary of findings 1](#).

Although not in the original protocol, these were prespecified for this update in previous versions of the review, and were not deviations from the information provided in the last updates [47, 50].

In this review update, we have added sensitivity analyses for survival outcomes to look at those studies that achieved higher rates of NMRD (> 40% at PCRS), reflecting a more modern surgical ethos of radicality, in response to criticisms of older studies.

Criteria for considering studies for this review

Types of studies

Randomised controlled trials (RCTs).

Types of participants

Women with advanced epithelial ovarian cancer (EOC) (International Federation of Gynecology and Obstetrics (FIGO) stage III/IV).

Types of interventions

Primary cytoreductive surgery (PCRS), with the aim of macroscopic resection or optimal cytoreduction (as defined by the study authors), followed by platinum-based chemotherapy, compared to platinum-based neoadjuvant chemotherapy (NACT) followed by interval CRS (ICRS), with the aim of resecting disease to the same degree as the PCRS group.

Outcome measures

We extracted data for direct outcome measures, relevant to patients and clinicians, including benefits, harms, and quality-of-life data, as detailed below.

Critical outcomes

- Overall survival: defined as death from any cause from time of randomisation.
- Progression-free survival: defined as time free of disease progression or death from time of randomisation.

Important outcomes

- Morbidity or adverse effects classified according to National Cancer Institute Common Terminology Criteria for Adverse Events (CTCAE) criteria [54]:
 - postoperative mortality;
 - surgically related adverse effects (grade 3+ (G3+)):
 - direct surgical morbidity (G3+) (e.g. bladder injury, intestinal obstruction, haematoma, local infection, duration of operation);
 - surgically related systemic morbidity (G3+) (e.g. deep vein thrombosis, pulmonary embolism, chest infection, cardiac events, haemorrhage/need for blood transfusion);
 - need for bowel resection, stoma formation and/or splenectomy;
 - recovery, including duration of hospital stay;
 - toxicity related to chemotherapy, grouped as haematological, gastrointestinal, genitourinary, skin, and neurological toxicity.
- Quality of life measured using a validated scale (e.g. EORTC QLQ-C30 [55, 56, 57], EORTC QLQ-OV28 [58]).
- Extent of CRS achieved (e.g. no NMRD, small volume residual disease (SVRD) (0.1 to 1 cm), large volume residual disease (LVRD) (larger than 1 cm) and combined macroscopic and SVRD (0.1 to ≤ 1 cm) (previously described as 'optimal')).

We present a summary of findings table reporting the following predefined outcomes.

- Overall survival
- Progression-free survival

- Postoperative mortality
- Surgically related adverse effects (grade 3+)
- Surgical resection: stoma formation
- Surgical resection: bowel resection
- EORTC QLQ-C30 at six months

Search methods for identification of studies

Electronic searches

We searched the following electronic databases on 9 October 2025, and for this update the additional search period was from 21 March 2024 onwards:

- Cochrane Central Register of Controlled Trials (CENTRAL; Issue 9, 2025) in the Cochrane Register of Studies Online (searched 9 October 2025) ([Supplementary material 1](#));
- MEDLINE Ovid (1966 to 9 October 2025) ([Supplementary material 1](#));
- Embase Ovid (1980 to 2025 week 41) ([Supplementary material 1](#)).

We formally searched the following trials registries from 13 March 2023 to 9 October 2025. Before April 2023, these were searched and relevant studies were updated monthly to CENTRAL by the Cochrane Gynaecological, Neuro-oncology and Orphan Cancer Review Group Information Specialist, so not searched separately.

- ClinicalTrials.gov (clinicaltrials.gov/)
- World Health Organization International Clinical Trials Registry Platform (WHO ICTRP) (trialsearch.who.int/)

Searching other resources

We searched the reference lists of relevant papers for further studies and contacted the authors of relevant studies to request information relating to their participation in unpublished studies. We sought papers in all languages and carried out translations as necessary.

We entered all relevant articles into PubMed, and using the 'related articles' feature, performed a further search for any other published articles. We also checked the reference lists of studies included in other systematic reviews identified by the search.

We did not separately handsearch conference abstracts, as abstracts for major conferences are published in journals encompassed by the search strategy (e.g. International Gynaecological Cancer Society; European Society of Medical Oncology; American Society of Clinical Oncology). We checked whether included studies had been retracted or were subject to corrections or expressions of concern.

Data collection and analysis

Selection of studies

Two review authors (a combination of CE, MS, SC, RB and JM for this update) independently selected trials from the results of the searches according to the inclusion criteria specified above. We resolved any disagreements by discussion and referral to a third review author, if required.

Data extraction and management

For this update, we replicated the data collection form for study characteristics and outcome data that had been piloted previously. Two review authors (CE and JM) independently extracted data. Any disagreements were resolved by consensus and discussion with a third review author (SC), if necessary. No attempt was made to blind review authors to authors of articles or to journals. Two review authors (CE and JM) checked and transferred study data into the analyses.

For each included study, we recorded details of trial methodology, study population and sample size, inclusion and exclusion criteria, intervention and comparison, duration of follow-up, and risk of bias. We extracted data relating to participant characteristics (age, performance status, ethnicity, histology, grade, extent of disease, previous therapies) and outcomes. For each outcome, we extracted the outcome definition and unit of measurement.

We extracted results as follows.

- For time-to-event data (survival and disease progression), we extracted the log of the hazard ratio (log (HR)) and its standard error. If these were not reported, we estimated the log (HR) and its standard error using Parmar's methods [59, 60, 61].
- For dichotomous outcomes (e.g. adverse events or deaths), we extracted the number of women in each treatment arm who experienced the outcome of interest and the number of women assessed at the endpoint to estimate a risk ratio.
- For continuous outcomes (e.g. quality-of-life measures), we extracted the final value and standard deviation (SD) of the

outcome of interest and the number of women assessed at the endpoint in each treatment arm to estimate the mean difference between treatment arms and its standard error.

Where data were missing or methods unclear, we contacted the study authors for further information. We entered data into Review Manager (RevMan) software [62], and five review authors (CE, MS, SC, AB, JM) checked the data for accuracy.

Risk of bias assessment in included studies

We assessed risk of bias in the included studies using Cochrane's Risk of Bias 1 tool (RoB 1), as per previous updates [63]. We assessed the following risk of bias domains:

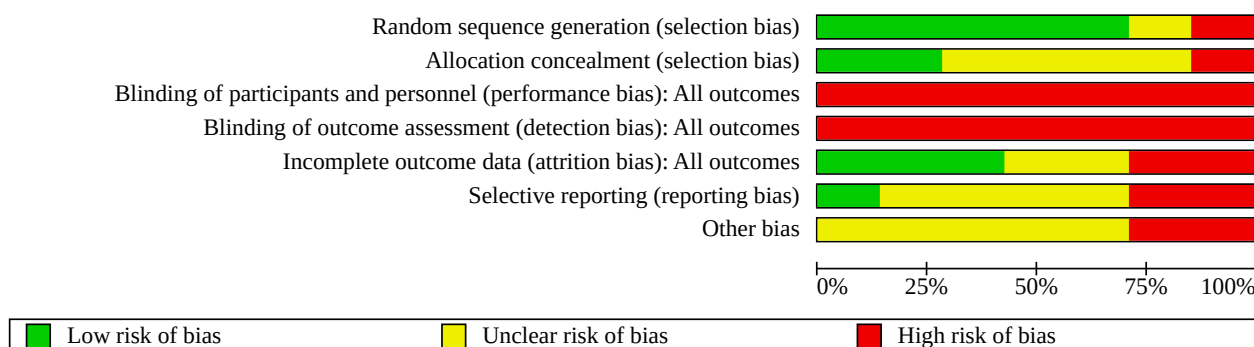
- random sequence generation and allocation concealment (selection bias);
- blinding of participants and assessors (performance and detection bias);
- incomplete outcome data (attrition bias);
- selective reporting of outcomes (reporting bias);
- other possible sources of bias.

Two review authors (a combination of CE, MS, SC, CC, and JM) independently applied the RoB 1 tool, with any differences of opinion resolved by discussion (Supplementary material 8). We summarised results in a risk of bias summary and graph (Figure 1; Figure 2). It is unlikely that lack of blinding could have affected overall survival or perioperative mortality, which is reflected in our GRADE assessment. However, other outcomes could have been influenced by lack of blinding, and this was considered accordingly.

Figure 1. Risk of bias summary: review authors' judgements about each risk of bias item for each included study

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment (detection bias): All outcomes	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Other bias
Abou-Taleb 2025	?	?	—	—	—	—	?
Chekman 2015	—	—	—	—	—	—	—
CHORUS	+	+	—	—	+	?	?
EORTC-55971	+	?	—	—	?	?	?
Onda 2016	+	+	—	—	+	+	—
SCORPION	+	?	—	—	?	?	?
TRUST	+	?	—	—	+	?	?

Figure 2. Risk of bias graph: review authors' judgements about each risk of bias item presented as percentages across all included studies



Measures of treatment effect

To measure the effect of treatment, we used the hazard ratio (HR) for time-to-event data; the risk ratio (RR) for dichotomous outcomes; and the mean difference (MD) between treatment arms for continuous outcomes.

Unit of analysis issues

No issues were noted.

Dealing with missing data

We noted levels of attrition. We did not impute missing outcome data for any outcome.

Reporting bias assessment

We did not produce funnel plots to assess the potential for small-study effects as we included only five studies in the meta-analysis.

Synthesis methods

If sufficient clinically similar studies were available, we pooled their adjusted results in meta-analyses.

- For time-to-event data, we pooled HRs using the generic inverse variance facility in RevMan.
- For dichotomous outcomes, we calculated RRs for each study, which we then pooled.
- For continuous outcomes, we pooled MDs between treatment arms at the end of follow-up, as all studies measured the outcome on the same scale; otherwise we would have pooled standardised mean differences (SMDs).

We used random-effects models with inverse variance weighting for all meta-analyses [64].

We assessed heterogeneity between studies by:

- visual inspection of forest plots;
- estimation of the percentage heterogeneity between studies that could not be ascribed to sampling variation [65];
- a formal statistical test of the significance of the heterogeneity [66]; and

- subgroup analyses where possible (see below). If there was evidence of substantial heterogeneity, we investigated and reported possible reasons.

Investigation of heterogeneity and subgroup analysis

As prespecified in previously published versions of the review, although not in the original protocol [67], we included the following subgroup analyses:

- age: younger than 50 years; younger than 60 years; 50 to 60 years; 50 to 70 years; 60 to 70 years; or over 70 years;
- extent of cytoreduction achieved: NMRD; SVRD (residual tumour deposits 1 cm or smaller); LVRD (residual tumour deposits larger than 1 cm);
- FIGO stage;
- ECOG performance status.

We evaluated these subgroup analyses with respect to our critical outcomes only.

Sensitivity analysis

In this review update, we performed sensitivity analyses for survival outcomes to select those studies that achieved higher rates of NMRD (> 40% at PCRS), reflecting a more modern surgical ethos of radicality, in response to criticisms of older studies. In future versions of this review, where possible, we will perform sensitivity analyses based on risk of bias (prioritising selection bias, then attrition bias, followed by reporting bias). We excluded from meta-analyses studies deemed at critical risk of bias (high risk in ≥ 4 domains).

Certainty of the evidence assessment

We presented the overall certainty of the evidence for each outcome ([Outcome measures](#)) according to the GRADE approach, which considers issues not only related to internal validity (risk of bias, inconsistency, imprecision, publication bias) but also to external validity such as directness of results [68]. We created a summary of findings table based on the methods described in the *Cochrane Handbook for Systematic Reviews of Interventions* [53] and using GRADEpro GDT software [69] (see [Summary of findings 1](#)). We used the GRADE checklist and GRADE Working Group certainty of evidence definitions [70]. We downgraded the evidence from 'high'

certainty by one level for serious (or by two levels for very serious) concerns for each limitation.

- High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.
- Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
- Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
- Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

Equity considerations

We did not investigate equity-related characteristics in this review, but did extract data on age and ethnicity, where this information was provided. We aimed to include anyone with ovarian cancer, regardless of gender identity.

Consumer involvement

Consumers were not involved in performing this review update, although the review authors did include patient-reported outcomes as part of the core review outcomes and key outcomes

were added, and the review text amended, following consumer review of all previous versions of this review.

RESULTS

Description of studies

Results of the search

For details of the search strategies, see [Supplementary material 1](#).

We identified a further 1994 titles and abstracts through our searches in this update (1756 after further de-duplication), in addition to the 3249 titles and abstracts identified in previous versions of this review. In total, seven RCTs of varying quality and size met the inclusion criteria. We identified one new completed study, and a previous ongoing study (published as an abstract only) and included them in this update. The studies assessed a total of 2650 women with stage III/IV ovarian cancer randomised to NACT followed by ICRS, or PCRS followed by chemotherapy.

[Figure 3](#) shows the screening process for this update with the number of studies brought forward from the previous versions [45, 46, 47, 48, 49, 50]. We identified two new references for one previously ongoing RCT (TRUST [71, 72, 73, 74]), and one new study from the full-text articles assessed for eligibility (Abou-Taleb 2025 [75, 76]).

Figure 3. Updated search to 9 Oct 2025 (from March 2024 in last update)

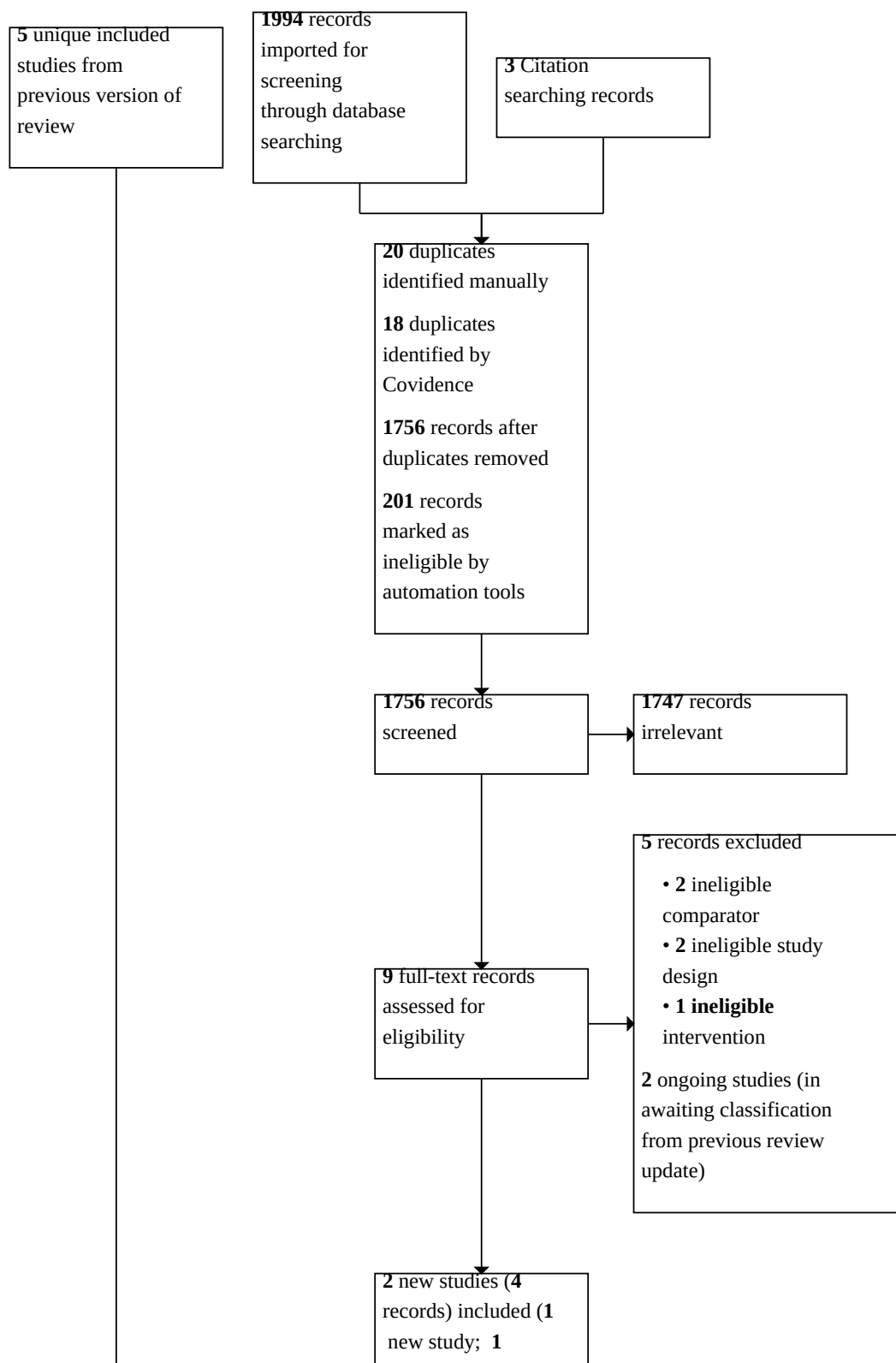
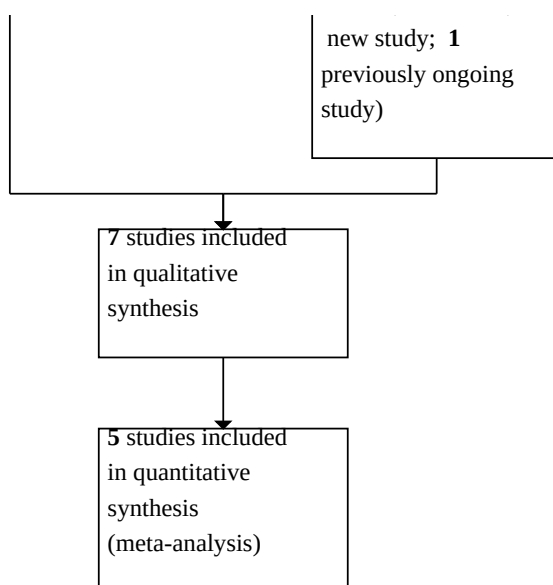


Figure 3. (Continued)



Included studies

We added two new studies to this review update. Seven studies met the eligibility criteria with a total of 2650 participants (2512 after post-randomisation exclusions). A summary of included studies and syntheses is provided in [Table 3](#); key study characteristics important for interpreting the syntheses and full details of the included studies are available in [Supplementary material 2](#). We were only able to include data from five studies in the meta-analyses (2380 women; CHORUS [77, 78, 79, 80, 81, 82, 83]; EORTC-55971 [43, 84, 85, 86, 87, 88, 89, 90]; Onda 2016 [91, 92, 93, 94, 95]; SCORPION [96, 97, 98, 99, 100, 101, 102]; TRUST). Data from Chekman 2015 [103, 104] and Abou-Taleb 2025 could not be included either due to the way in which they were reported, because they were at high risk of bias, or both.

Characteristics of studies

The included studies were conducted in Algeria, Argentina, Austria, Belgium, Canada, Denmark, Egypt, France, Germany, Ireland, Italy, Japan, the Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, the UK, and the USA. Chekman 2015 was conducted in a single centre in Algeria and Abou-Taleb 2025 in a single centre in Egypt. The included studies were published between 2010 and 2025. All studies included women with presumed stage III-IV EOC, with one including a few participants with stage IIB (Abou-Taleb 2025). Four studies were multicentre studies (CHORUS; EORTC-55971; Onda 2016; TRUST), and three were single-institution studies (Abou-Taleb 2025; Chekman 2015; SCORPION), one of which had all CRS performed by a single surgeon (Chekman 2015). We thank the authors of all included studies for responding to requests for further information at different time points during the course of this review and updates (Chekman 2015; CHORUS; EORTC-55971; Onda 2016; SCORPION; TRUST), and look forward to receiving further data from the authors of the study published in abstract following peer-reviewed publication.

Characteristics of participants

Four studies made a clinical diagnosis of stage IIIC-IV EOC based on imaging and tumour markers (Abou-Taleb 2025; CHORUS; EORTC-55971; Onda 2016). One study performed diagnostic laparoscopy on all but three participants to confirm the diagnosis (Chekman 2015). Four of these studies used the 1988 FIGO staging system (see [Table 1](#)). SCORPION and TRUST used the 2014 FIGO staging system (see [Table 2](#)). It is not clear what FIGO staging system Abou-Taleb 2025 used, however, given the timing of the study, it is expected they used the 2014 parameters.

One study required all potential participants to undergo a staging laparoscopy in order to be eligible for randomisation to the study arms (SCORPION). This was to obtain histology and confirm diagnosis, as well as to assess the tumour load. Tumour load was assessed using a predictive index (PI). Only participants with a PI score between 8 and 12, corresponding to a moderate-to-high tumour load, were eligible for randomisation. If it was deemed not possible to perform a staging laparoscopy due to large masses occupying the abdominal cavity infiltrating the abdominal wall or the presence of mesenteric retraction, participants were withdrawn from the study. A total of 225 participants underwent a staging laparoscopy, but only 171 went on to be randomised. Participants were significantly younger and had better performance status compared with the other studies that provided data. TRUST required all those randomised to NACT to have histological confirmation, and participants in both arms were withdrawn if they did not meet the histological or staging eligibility criteria, either after biopsy or at surgery. To be eligible, participants also had to have disease thought to be amenable to macroscopic resection.

Treatment format

Five studies aimed to give six cycles of chemotherapy in each arm, starting either before or after CRS (Chekman 2015; SCORPION; CHORUS; EORTC-55971; TRUST). One of these studies primarily gave six cycles of chemotherapy prior to CRS (Chekman 2015),

which many would term delayed CRS, rather than ICRS. The other four studies aimed to perform surgery after three (CHORUS; EORTC-55971; TRUST), or three or four cycles of NACT (SCORPION). The sixth study aimed to give eight cycles of chemotherapy, with those in the ICRS arm offered surgery after four cycles (Onda 2016). Those in the PCRS arm were also reassessed, and 49 women (32.9%) were offered a further CRS procedure in addition to PCRS. Abou-Taleb 2025 did not detail the number of cycles of chemotherapy received by participants.

Chemotherapy regimens were platinum-based (carboplatin and paclitaxel every three weeks in the majority of studies). In three studies (SCORPION; TRUST; Abou-Taleb 2025), bevacizumab, in combination with standard chemotherapy and as maintenance therapy, was offered to selected participants. TRUST also offered Poly (ADP-ribose) polymerase (PARP) inhibitors as maintenance therapy to selected participants.

Outcome reporting and assumptions required for synthesis

Surgical complications

In all the included studies, not all participants went on to have CRS, with rates higher in the NACT treatment arm, largely due to progressive disease on NACT. Surgical complication rates were therefore based on those who did undergo surgery, rather than the whole intention-to-treat cohort. Surgical complications in one study were reported for an initial cohort (SCORPION), powered for the co-primary endpoint of major perioperative morbidity. Further participants were recruited to attain statistical power on progression-free survival, and surgical complications were only reported for the initial 110 cohort.

All studies reported the degree of residual disease remaining following CRS (NMRD; SVRD; LVRD), although the definition of 'optimal' CRS has changed over the time period in which the studies were conducted.

Survival outcomes

All studies reported progression-free survival and overall survival, although some studies published overall survival data at later time points from the original publications. Two studies were in abstract form only (Chekman 2015; TRUST); the authors kindly provided us with more information on request. Unfortunately, we were unable to analyse the survival outcomes for Chekman 2015, as they did not provide data for time-to-event outcomes in a format that was appropriate for inclusion.

Adverse event data

Five studies reported data using CTCAE tools (versions 2.0 and 3.0; CHORUS; EORTC-55971; Onda 2016; SCORPION; TRUST).

Quality-of-life data

Four studies reported quality-of-life data using the validated EORTC QLQ-C30 and EORTC QLQ-OV28 questionnaires (CHORUS; EORTC-55971; SCORPION; TRUST), although two of these studies reported this information at later time points than the initial survival or surgical data, or both (EORTC-55971; SCORPION), and more detailed data from TRUST will be required to allow inclusion in the meta-analysis.

Excluded studies

We excluded five new studies in this update, adding to the 115 excluded in previous versions, for a total of 120 excluded studies. Reasons for exclusion are documented in [Supplementary material 3](#). Common reasons for exclusion across all versions of this review included non-randomised study design, lack of surgical comparison arm, RCT of further surgery after PCRS versus no further surgery, RCTs comparing different numbers of cycles of chemotherapy prior to ICRS, and systematic reviews of RCTs and NRSs.

Liu 2004 [105], an RCT that compared NACT plus iliac artery embolisation versus PCRS, was originally included in the 2006 version of this review. In a previous review update, we revised our assessment of this study and excluded it, as the study findings may have been attributable to NACT versus PCRS, iliac artery embolisation, or a combination of the two, because NACT versus PCRS was not the only variable in the study, as iliac artery embolisation was not delivered in both arms.

Ongoing studies

We identified one new ongoing study (IRCT20250902067090N1 2025 [106]), for a total of six ongoing studies (Kumar 2009 [107, 108, 109, 110, 111, 112, 113]; NCT04257786 [114]; NCT04515602 [115]; NCT05371301 [116, 117]; SUNNY [118, 119, 120]).

Kumar 2009 had reported interim analyses in abstract form, but the outcomes were inadequately reported, and the risk of bias profile was unclear, so we briefly discussed this trial in [Agreements and disagreements with other studies or reviews](#) in the [Discussion](#) and added it to the list of ongoing studies (see [Supplementary material 5](#)). Unfortunately, despite contact with the author, no further data have been provided to date for inclusion in the review.

Studies awaiting classification

We categorised no new studies as awaiting classification, with four studies in this category in the previous version (ChiCTR2000035918 2020 [121]; ChiCTR2100047852 2021 [122]; Jiang 2020 [123, 124]; Wen 2021 [125]) (see [Supplementary material 4](#)). In all of these studies, the trial design was unclear due to inconsistencies in their description or lack of information, or both; we have contacted the authors for clarification. We identified an additional reference to Jiang 2020.

Risk of bias in included studies

For this update of the review, a combination of two out of five review authors (from CE, SC, JM, MS, and CC) independently reassessed the risk of bias in each included study according to predefined criteria stated in the [Methods](#) section ([Figure 1](#)). For details on the individual studies, see [Supplementary material 2](#).

We assessed the Chekman 2015 study at high risk in all domains. Ninety women with FIGO stage IIIC ovarian carcinoma were enrolled and underwent surgery, but only 82 women were randomised, 41 to PCRS/chemotherapy and 41 to NACT/ICRS. The randomisation was performed in the operating room by "random draw" by someone other than the surgeon, once verification of inclusion criteria and resectability under laparoscopy or laparotomy had been confirmed. For these reasons, we have not included data from the study in the results or the analyses; details of the study can be found in [Supplementary material 2](#).

We assessed the Abou-Taleb 2025 study as high risk in four of the seven domains, whilst it was unclear risk in the remaining three domains. Not all participants were analysed on an intention-to-treat basis. There was also no documentation detailing attrition of participants. For these reasons, we have not included data from the study in the results or the analyses; details of the study can be found in [Supplementary material 2](#)

Selection bias

Random sequence generation

We assessed risk of bias for random sequence generation as low in five studies because the randomisation was performed centrally (CHORUS; EORTC-55971; Onda 2016; SCORPION; TRUST).

Allocation concealment

We assessed SCORPION at unclear risk of bias for allocation concealment because of the complex inclusion requirements and highly selected population. Women were only eligible for randomisation into the study once they had undergone a staging laparoscopy to assess disease burden. The staging laparoscopy was used as a triage tool to assess eligibility for the study. If a staging laparoscopy was unfeasible, women were removed from the study. If the staging laparoscopy was successful, a PI value was calculated based upon seven parameters: presence or absence of omental cake, extensive carcinomatosis of the peritoneal or diaphragmatic surfaces, mesenteric retraction, infiltration of the stomach, spleen, or bowel, and/or superficial liver metastases. The PI scoring system was based upon earlier work by the same group [126, 127, 128]. Participants were included if their PI score was between 8 and 12. This was chosen to include participants in the area of equipoise, with a relatively high tumour load, related to lower chances of optimal cytoreduction and worse prognosis than those who had lower scores (and deemed suitable to offer PCRS), but not so high (> 12) that the likelihood of achieving NMRD after surgery was thought to be much lower. These complexities in trial design introduce potential sources of bias and may limit applicability to the general advanced ovarian cancer population.

In the EORTC-55971 study, data from 48 women from Argentina were excluded owing to "potential authorisation irregularities"; however, the investigators stated that their results were similar when these excluded data were included. This study is therefore at unclear risk of bias for allocation concealment.

We assessed the TRUST study at unclear risk of bias for allocation concealment as, following randomisation, the outcome was faxed from the co-ordinating centre in Germany.

Blinding of participants and assessors (performance and detection bias)

All studies were open-label, and outcome assessment was not blinded. This is probably not an issue for our critical outcomes (i.e. overall survival and postoperative mortality); however, it could lead to performance and detection bias with regard to other outcomes or subgroups (e.g. extent of cytoreduction achieved). The importance of blinding of outcome assessment in ovarian cancer studies has been raised in a Gynecologic Cancer InterGroup (GCIg) consensus statement [20]. Data for these latter outcomes should thus be interpreted with caution; we deemed all studies to be at high risk of bias.

Incomplete outcome data (attrition bias)

As discussed above, EORTC-55971 excluded data from 48 women from Argentina. The exclusions appeared erroneously as pre-randomisation exclusions on the published study flow diagram. The study was therefore at unclear risk of attrition bias.

Selective reporting of outcomes (reporting bias)

Not all outcomes for all participants were reported in CHORUS, EORTC-55971, and SCORPION, as discussed here and in other sections. In SCORPION, the final trial cohort consisted of 171 women. An early publication included 110 women and reported short-term quality-of-life outcomes and surgical outcomes. TRUST reported main outcomes, however, the reporting of outcomes is incomplete at present as only an abstract has been published.

Other potential sources of bias

Supplementary data in CHORUS table 7 show that hysterectomy/bilateral salpingo-oophorectomy and omentectomy were performed in varying proportions in the different arms. It is unclear what effect this might have had on outcomes and could be a potential source of bias. In the Onda 2016 study, 14 women (one in the PCRS group and 13 in the NACT group) underwent some type of additional surgery (off-protocol treatment). These off-protocol operations were not included as PCRS or ICPS in the analysis. There appeared to be more off-protocol surgery in the NACT group. No intention-to-treat analysis was performed. These issues could be another potential source of bias.

In the original 110-participant cohort in SCORPION, all 55 women in the PCRS arm had upper abdominal surgical procedures performed compared to 22 of 52 women who underwent ICPS (42.3%). The median duration of entire treatment from randomisation to completion of medical treatment was also longer in the PCRS arm (38 weeks versus 28 weeks). This was due to an almost two-week difference in time to start post-surgery chemotherapy (median time post-PCRS 40 days; median time post-ICPS 27 days; $P < 0.001$). The short-term surgical outcomes for the additional 61 women were not included in the later publications of overall survival outcomes and long-term detailed quality-of-life outcomes.

Synthesis of results

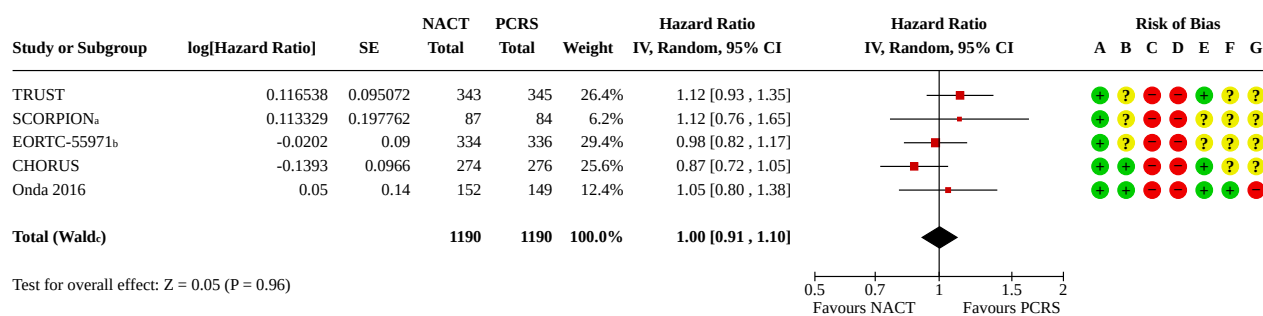
A complete record of all comparisons and analyses is available for this review in [Supplementary material 6](#), and a full data package is available in [Supplementary material 7](#).

Critical outcomes

Overall survival (Analyses 1.1 to 1.4)

Meta-analysis of five studies (CHORUS; EORTC-55971; Onda 2016; SCORPION; TRUST), assessing 2380 women, demonstrated little or no difference in overall survival between NACT and PCRS for initial treatment in advanced ovarian cancer (HR 1.00, 95% CI 0.91 to 1.10; $I^2 = 2\%$; 5 studies, 2380 participants; high-certainty evidence; [Figure 4](#); Analysis 1.1). Survival curves for studies demonstrated non-proportionality with increased early deaths in the PCRS groups in some studies (EORTC-55971; SCORPION; TRUST). Post hoc sensitivity analysis, only including those studies with high rates of NMRD after PCRS (> 40%), did not change the outcome (HR 1.12, 95% CI 0.95 to 1.33; $I^2 = 0\%$; 2 studies; 859 participants).

Figure 4.

**Footnotes**

^aAuthors informed us that PCRS was the reference group for HR, although due to evidence of non-proportionality of hazards visible on Kaplan-Meier curve and overlapping, similar survival c

^bWe have applied 95% CIs (investigators reported 90% CIs).

^cCI calculated by Wald-type method.

^d Tau^2 calculated by DerSimonian and Laird method.

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

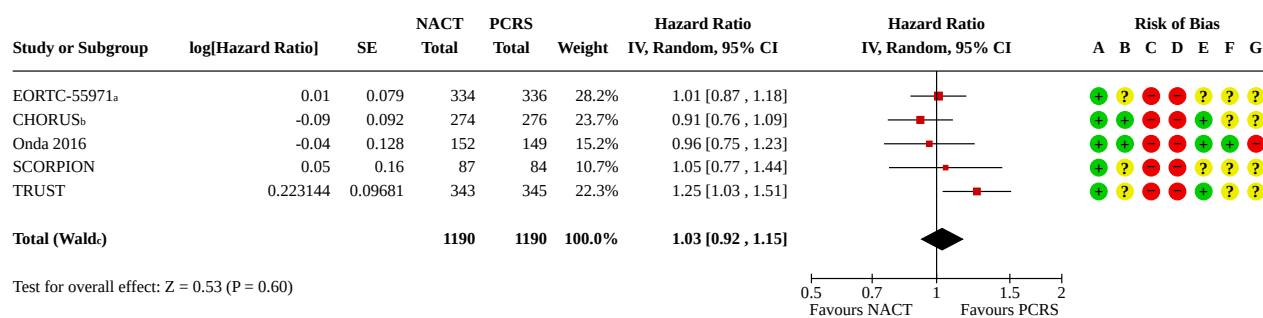
On subgroup analysis, there was no evidence that age, stage, or extent of residual disease was an effect modifier (Analysis 1.2; Analysis 1.3; Analysis 1.4). Survival data by stage were not available for one study (SCORPION), and there were only 8 and 13 women with stage IV disease in the NACT and PCRS arms, respectively.

Meta-analysis of CHORUS and EORTC-55971 showed that survival by performance status did not differ, although CIs were wider for those with ECOG performance status of 2+, due to fewer participants in this subgroup (performance status 0: HR 0.97, 95% CI 0.79 to 1.19; performance status 1: HR 0.96, 95% CI 0.81 to 1.15; performance status 2+ HR 0.97, 95% CI 0.71 to 1.29; 2 studies). We did not formally analyse overall survival by performance status, as this was not prespecified.

Progression-free survival (Analysis 1.5)

Meta-analysis of five studies (CHORUS; EORTC-55971; Onda 2016; SCORPION; TRUST), assessing 2380 women, found likely little or no difference in risk of disease progression between NACT and PCRS for initial treatment in advanced ovarian cancer (HR 1.03, 95% CI 0.92 to 1.15; $I^2 = 34\%$; 5 studies, 2380 participants; moderate-certainty evidence; Figure 5; Analysis 1.5). Post hoc sensitivity analysis, only including those studies with high rates of NMRD after PCRS ($> 40\%$), suggests that there may be a small improvement in progression-free survival, although this outcome was not prespecified (HR 1.19, 95% CI 1.01 to 1.40; $I^2 = 0\%$; 2 studies; 859 participants).

Figure 5.

**Footnotes**^aWe have applied 95% CIs (Investigators used 90% CIs)^b0.09^cCI calculated by Wald-type method.^d Tau^2 calculated by DerSimonian and Laird method.**Risk of bias legend**

(A) Random sequence generation (selection bias)

(B) Allocation concealment (selection bias)

(C) Blinding of participants and personnel (performance bias)

(D) Blinding of outcome assessment (detection bias)

(E) Incomplete outcome data (attrition bias)

(F) Selective reporting (reporting bias)

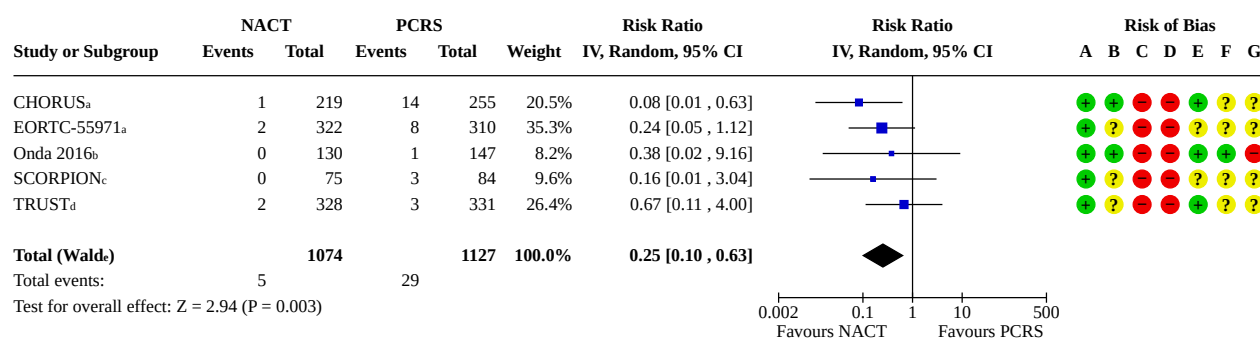
(G) Other bias

Important outcomes**Postoperative mortality (Analysis 1.6)**

Meta-analysis of five studies (CHORUS; EORTC-55971; Onda 2016; SCORPION; TRUST), assessing the 2201 women who had surgery, found that women in the NACT arm had less risk of postoperative mortality than those in the PCRS arm (RR 0.25, 95% CI 0.10 to 0.63; $I^2 = 0\%$; 5 studies, 2201 participants; high-certainty evidence; [Figure 6](#); Analysis 1.6). Five of 1074 (0.5%) women died within a

month of surgery in the NACT arm compared to 29 of 1127 (2.6%) women in the PCRS arm. Overall, 33 of 1127 (2.9%) women died due to postoperative complications, as there were four additional deaths in SCORPION due to postoperative complications in women who survived more than 30 days after surgery. Their deaths were attributed to postoperative complications and delayed to more than 30 days due to high-dependency care. Overall postoperative mortality was therefore 2.9% (33/1127) in the PCRS group versus 0.5% (5/1074) in the NACT group (Analysis 1.6).

Figure 6.

**Footnotes**^adeaths within 28 days of surgery^bDefined as 'treatment-related deaths related to surgery' within 4 weeks of surgery^cFagotti 2016 includes 3 post-op deaths within 30 days; there were a further 4 late post-op deaths, over 30 days, which were directly surgically related and due to post-op complica^ddeaths within 30 days of surgery^eCI calculated by Wald-type method.^fTau² calculated by DerSimonian and Laird method.**Risk of bias legend**

(A) Random sequence generation (selection bias)

(B) Allocation concealment (selection bias)

(C) Blinding of participants and personnel (performance bias)

(D) Blinding of outcome assessment (detection bias)

(E) Incomplete outcome data (attrition bias)

(F) Selective reporting (reporting bias)

(G) Other bias

Surgically related adverse effects (grade 3+) (Analysis 1.7)

These are defined as G3+ events related to the surgical procedure within 30 days of surgery. SCORPION reported major perioperative morbidity, initially when the study had randomised 110 women. Some level of granularity in adverse events was not given in the follow-up publication, so some analyses were based on the initial cohort (n = 110), whereas analyses included in the follow-up publication included all 171 women.

Some studies reported all adverse events grade 3 or worse (G3+ AEs) during the study period (CHORUS; EORTC-55971; Onda 2016), whereas one study reported surgically related G3+ AEs only (SCORPION). TRUST reported adverse events within 28 days of surgery, but severity data are not yet available. There were probably clinically meaningful differences in favour of NACT compared to PCRS with regard to overall surgically related adverse effects (G3+): 9.4% in the NACT group versus 22.2% in the PCRS group (RR 0.34, 95% CI 0.14 to 0.82; I² = 82%; 3 studies, 1094 participants; Figure 7; moderate-certainty evidence; Analysis 1.7.1).

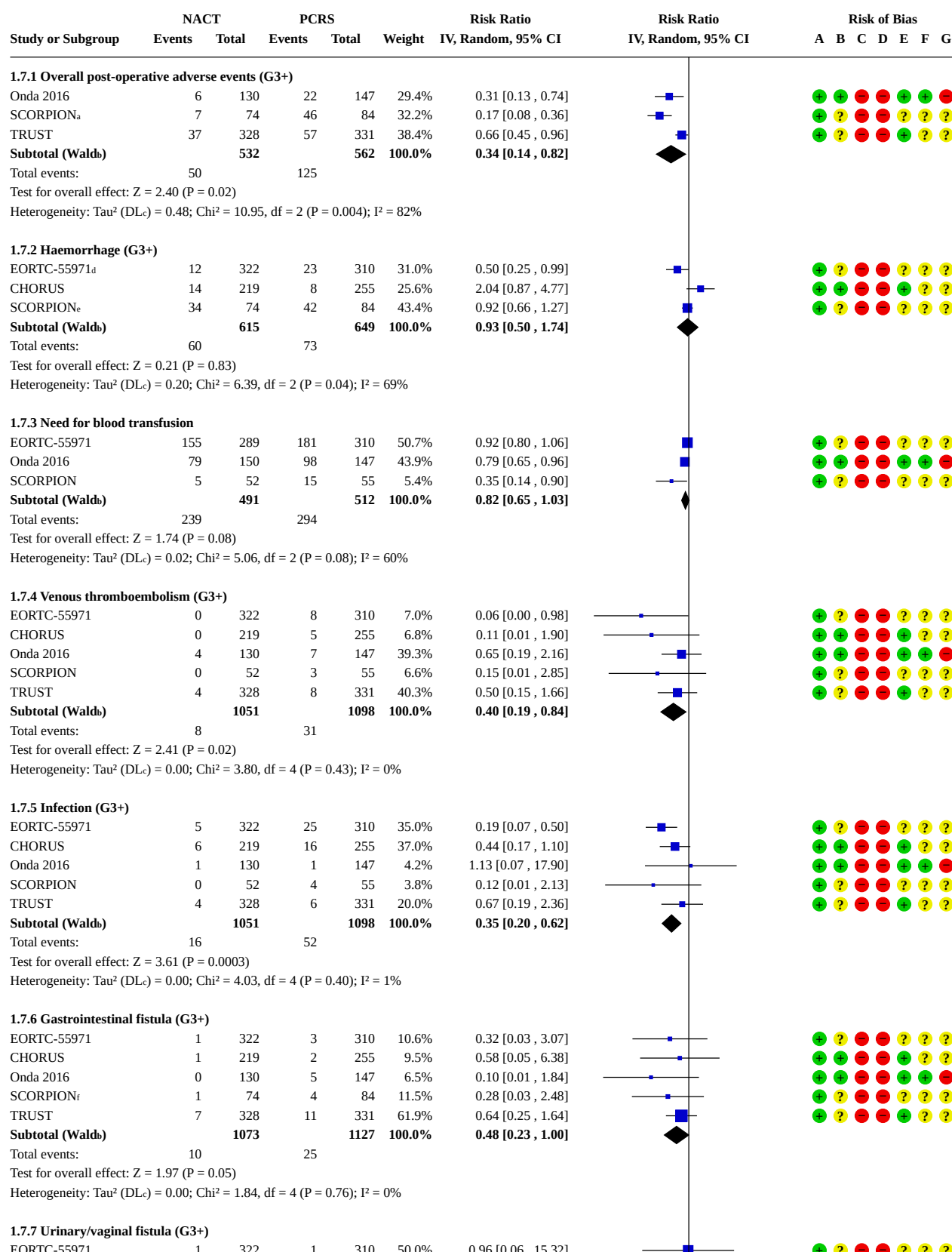
Figure 7.


Figure 7. (Continued)

1.7.7 Urinary/vaginal fistula (G3+)

EORTC-55971	1	322	1	310	50.0%	0.96 [0.06 , 15.32]
CHORUS	1	219	1	255	50.0%	1.16 [0.07 , 18.51]
Subtotal (Wald)		541		565	100.0%	1.06 [0.15 , 7.49]

Total events:

2

2

Test for overall effect: $Z = 0.06$ ($P = 0.95$)

Heterogeneity: Tau^2 (DL_0) = 0.00; $\text{Chi}^2 = 0.01$, $\text{df} = 1$ ($P = 0.92$); $I^2 = 0\%$

1.7.8 Nausea (G3+)

CHORUS	1	219	12	255	52.1%	0.10 [0.01 , 0.74]
SCORPION	2	51	1	52	47.9%	2.04 [0.19 , 21.80]
Subtotal (Wald)		270		307	100.0%	0.42 [0.02 , 8.23]

Total events:

3

13

Test for overall effect: $Z = 0.57$ ($P = 0.57$)

Heterogeneity: Tau^2 (DL_0) = 3.37; $\text{Chi}^2 = 3.66$, $\text{df} = 1$ ($P = 0.06$); $I^2 = 73\%$

1.7.9 Vomiting (G3+)

CHORUS	1	219	12	255	48.2%	0.10 [0.01 , 0.74]
SCORPION	3	51	2	52	51.8%	1.53 [0.27 , 8.77]
Subtotal (Wald)		270		307	100.0%	0.41 [0.03 , 6.03]

Total events:

4

14

Test for overall effect: $Z = 0.66$ ($P = 0.51$)

Heterogeneity: Tau^2 (DL_0) = 2.87; $\text{Chi}^2 = 4.07$, $\text{df} = 1$ ($P = 0.04$); $I^2 = 75\%$

1.7.10 Diarrhoea (G3+)

CHORUS	2	219	4	255	100.0%	0.58 [0.11 , 3.15]
Subtotal		219		255	100.0%	0.58 [0.11 , 3.15]

Total events:

2

4

Test for overall effect: $Z = 0.63$ ($P = 0.53$)

Heterogeneity: Not applicable

1.7.11 Neutropenia (G3+)

SCORPION	9	51	8	52	100.0%	1.15 [0.48 , 2.74]
Subtotal		51		52	100.0%	1.15 [0.48 , 2.74]

Total events:

9

8

Test for overall effect: $Z = 0.31$ ($P = 0.76$)

Heterogeneity: Not applicable

1.7.12 Neutrotoxicity (G3+)

SCORPION	2	51	2	52	100.0%	1.02 [0.15 , 6.97]
Subtotal		51		52	100.0%	1.02 [0.15 , 6.97]

Total events:

2

2

Test for overall effect: $Z = 0.02$ ($P = 0.98$)

Heterogeneity: Not applicable

1.7.13 Thrombocytopenia (G3+)

SCORPION	2	51	0	52	100.0%	5.10 [0.25 , 103.61]
Subtotal		51		52	100.0%	5.10 [0.25 , 103.61]

Total events:

2

0

Test for overall effect: $Z = 1.06$ ($P = 0.29$)

Heterogeneity: Not applicable

1.7.14 Febrile neutropenia (G3+)

SCORPION	1	51	0	52	100.0%	3.06 [0.13 , 73.36]
Subtotal		51		52	100.0%	3.06 [0.13 , 73.36]

Total events:

1

0

Test for overall effect: $Z = 0.69$ ($P = 0.49$)

Heterogeneity: Not applicable

1.7.15 Renal toxicity (G3+)

SCORPION	0	51	0	52		Not estimable
Subtotal		51		52		Not estimable

Total events:

0

0

Test for overall effect: Not applicable

Heterogeneity: Not applicable

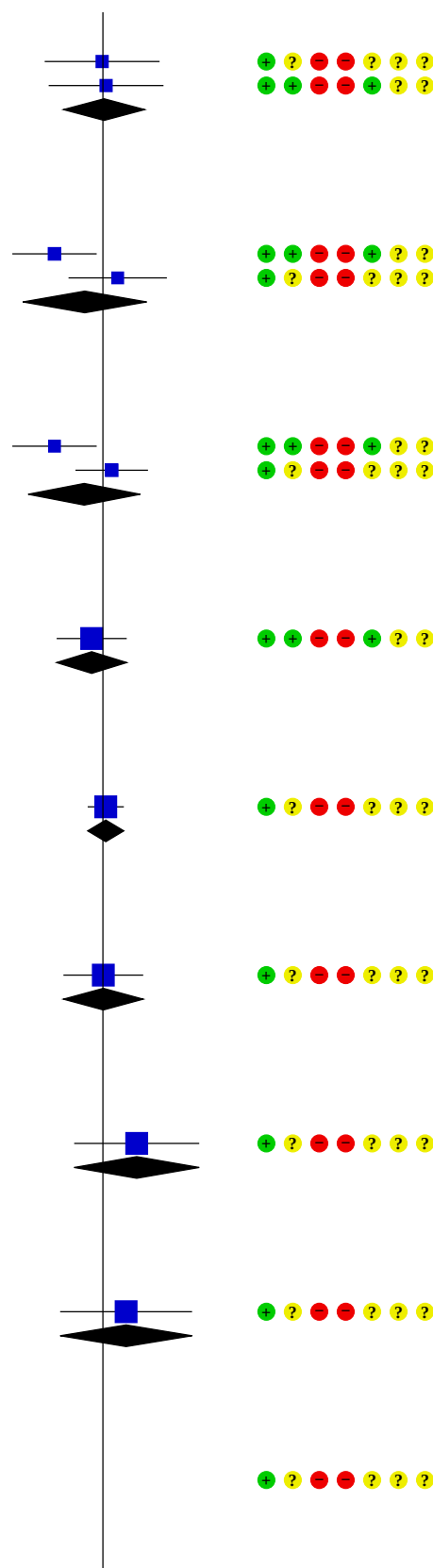


Figure 7. (Continued)



Footnotes

^awithin 30 days of surgery. Further post-op G3+ AE > 30 days (NACT = 1; PCRS = 13)

^bCI calculated by Wald-type method.

^cTau² calculated by DerSimonian and Laird method.

^dResults for all G3+ AEs in this trial are per protocol, not ITT.

^eEstimated Blood Loss >750 ml for those who had surgery

^fthree pancreatic fistulae and one biliary fistula

Risk of bias legend

(A) Random sequence generation (selection bias)

(B) Allocation concealment (selection bias)

(C) Blinding of participants and personnel (performance bias)

(D) Blinding of outcome assessment (detection bias)

(E) Incomplete outcome data (attrition bias)

(F) Selective reporting (reporting bias)

(G) Other bias

The following G3+ AEs were reported (Analysis 1.7.2 to Analysis 1.7.15) [54].

Haemorrhage (G3+) and blood transfusion requirements (Analyses 1.7.2 and 1.7.3)

Meta-analysis of three studies (CHORUS; EORTC-55971; SCORPION), assessing 1264 women, found there may be little to no difference in risk of haemorrhage (G3+) between NACT and PCRS for initial treatment in advanced ovarian cancer (RR 0.93, 95% CI 0.50 to 1.74; $I^2 = 69\%$; 3 studies, 1264 participants; Figure 7; Analysis 1.7.2). Haemorrhage is defined by the National Cancer Institute CTCAE criteria (G3+ $\geq$ 3-unit blood transfusion).

In the CHORUS and EORTC-55971 studies, the need for blood transfusions and average blood loss were not reported in the published versions of the studies. TRUST provided unpublished data on patients who received more than 10 packed red blood cells within 24 hours of their operation only. However, EORTC-55971 provided unpublished data with respect to the number of women who received blood transfusions in the NACT and PCRS groups. Meta-analysis of three trials (SCORPION; Onda 2016; EORTC-55971), assessing 1085 women, suggested that NACT and PCRS may result in a slight reduction in need for a blood transfusion after surgery compared to PCRS (RR 0.82, 95% CI 0.65 to 1.03; $I^2 = 60\%$; 3 studies, 1003 participants; Figure 7; Analysis 1.7.3).

Venous thromboembolism (G3+) (Analysis 1.7.4)

Meta-analysis of data from five studies found there may be a reduction in risk of venous thromboembolism in the NACT arm versus the PCRS arm (CHORUS; EORTC-55971; Onda 2016; SCORPION; TRUST), although this was based on a low number of events ($n = 39$) (RR 0.40, 95% CI 0.19 to 0.84; $I^2 = 0\%$; 5 studies, 2149 participants; Figure 7; Analysis 1.7.4).

Infection (G3+) (Analysis 1.7.5)

There is probably a small reduction in risk of infection in the NACT arm compared with the PCRS arm (RR 0.35, 95% CI 0.20 to 0.62; $I^2 = 1\%$; 5 studies, 2149 participants; Figure 7; Analysis 1.7.5).

Other adverse effects (G3+) (Analyses 1.7.6 to 1.7.15)

Meta-analysis of data from four studies found there may be little to no difference between groups for risk of urinary/vaginal fistula, nausea, vomiting, diarrhoea, neutropenia, neurotoxicity, thrombocytopenia, febrile neutropenia, and renal toxicity (see Analysis 1.7.6 to Analysis 1.7.15; CHORUS; EORTC-55971; Onda 2016; SCORPION). From meta-analysis of five studies, there may be a reduction in gastrointestinal fistula in the NACT arm compared with the PCRS arm (RR 0.48, 95% CI 0.23 to 1.00; $I^2 = 0\%$; 5 studies, 2200 participants; Figure 7; Analysis 1.7.6).

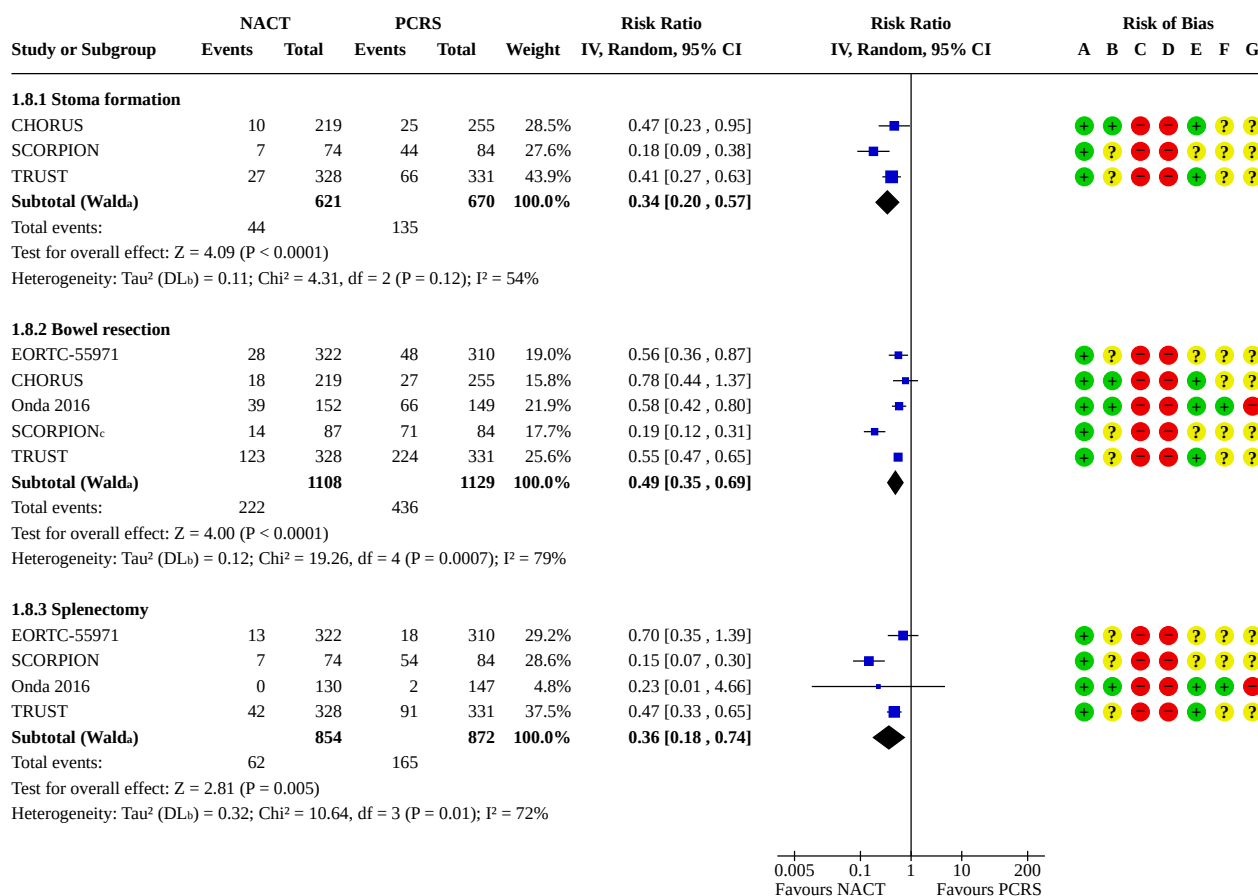
Surgical resection (Analysis 1.8)

See Analysis 1.8 for details.

Stoma formation (Analysis 1.8.1)

There was probably a large reduction in rate of stoma formation (colostomy or ileostomy) in the NACT arm versus the PCRS arm (RR 0.34, 95% CI 0.20 to 0.57; $I^2 = 54\%$; 3 studies, 1291 participants; Figure 8; moderate-certainty evidence; Analysis 1.8.1), although only three studies presented data (SCORPION; CHORUS; TRUST). Rates of stoma formation were 44/621 in the NACT arm (7.1%) and 135/670 in the PCRS arm (20.1%), representing a 2.8-fold difference in the risk of requiring a stoma following surgery. The higher rates of stoma formation in SCORPION and TRUST (17.6% versus 7.4% in CHORUS) likely reflect the increase in surgical radicality for ovarian cancer surgery over time, lower levels of visible disease after surgery, and the exclusion of individuals with low or very high PI scores. On post-hoc sensitivity analysis, including only those studies with high NMRD rates, the differences were 8.5% versus 26.5% for NACT and PCRS, respectively (RR 0.29, 95% CI 0.13 to 0.64; $I^2 = 73\%$; 2 studies 817 participants).

Figure 8.



Footnotes

^aCI calculated by Wald-type method.^b Tau^2 calculated by DerSimonian and Laird method.^cSingle bowel resection (NACT = 10 versus PCRS = 52); multiple bowel resections (NACT = 4 versus PCRS = 19)

Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Blinding of outcome assessment (detection bias)
- (E) Incomplete outcome data (attrition bias)
- (F) Selective reporting (reporting bias)
- (G) Other bias

Bowel resection (Analysis 1.8.2)

Based on data from five studies, women in the NACT arm were probably less likely to require a bowel resection (large and small bowel data combined) than women in the PCRS arm (RR 0.49, 95% CI 0.35 to 0.69; $I^2 = 79\%$; 5 studies, 2237 participants; Figure 8; moderate-certainty evidence; Analysis 1.8.2; CHORUS; EORTC-55971; Onda 2016; SCORPION; TRUST). Rates in the NACT arm were 222/1108 (19.6%) compared with 38.6% in the PCRS arm, representing an almost two-fold increased risk in the PCRS arm. On post-hoc sensitivity analysis, including only those studies with high NMRD rates, the differences were 33.0% versus 71.1% for NACT and PCRS, respectively (RR 0.33, 95%CI 0.12 to 0.95; $I^2 = 94\%$; 2 studies 817 participants).

Splenectomy (Analysis 1.8.3)

Based on data from four studies, women in the NACT arm are probably less likely to require a splenectomy than women in the PCRS arm (RR 0.36, 95% CI 0.18 to 0.74; $I^2 = 72\%$; 4 studies, 1726 participants; Figure 8; Analysis 1.8.3; EORTC-55971; Onda 2016; SCORPION; TRUST). On post-hoc sensitivity analysis, including only those studies with high NMRD rates, the differences were even more apparent (12.2% versus 34.9% for NACT and PCRS, respectively: RR 0.27, 95%CI 0.09 to 0.85; $I^2 = 88\%$; 2 studies 817 participants).

Quality of life (Analyses 1.9 to 1.14)

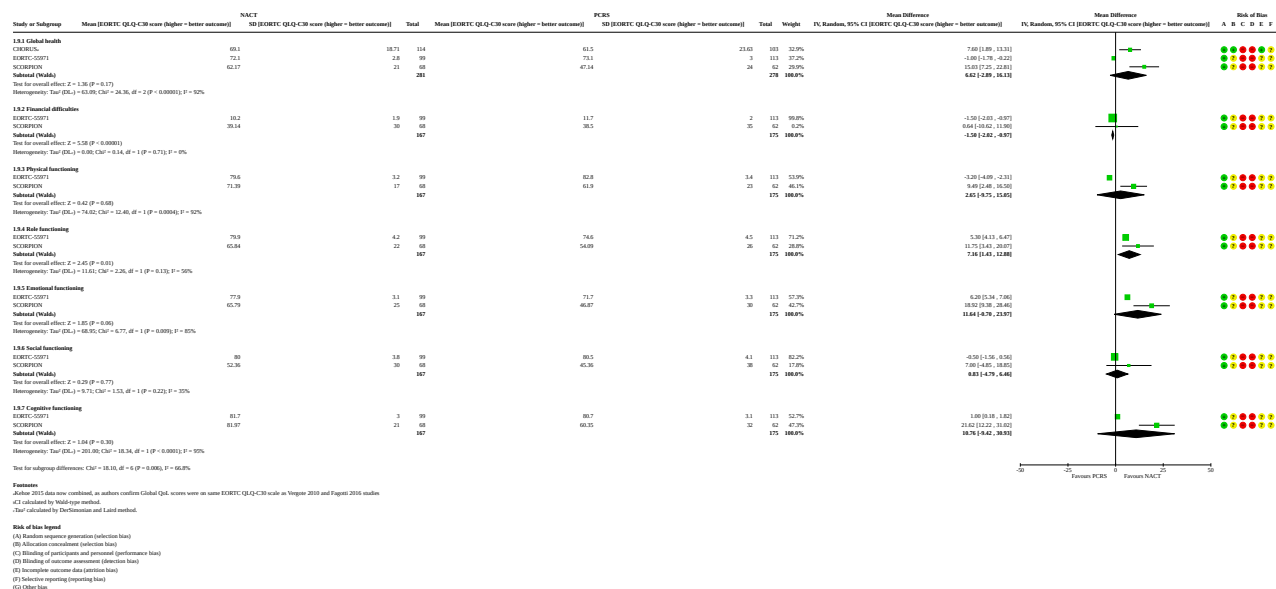
See Supplementary material 6 for Analyses 1.9 to 1.14.

Three studies, assessing 559 women, reported on quality of life (QoL) at 6 and 12 months (only 397 women) using the EORTC

QLQ-C30 (CHORUS; EORTC-55971; SCORPION). The EORTC QLQ-C30 produced imprecise results in three studies, with high levels of heterogeneity (global QoL at 6 months: MD 6.62, 95% CI –2.89 to 16.13; $I^2 = 92\%$; 3 studies, 559 participants; [Figure 9](#); very low-

certainty evidence; Analysis 1.9.1; and QoL at 12 months: MD 1.78, 95% CI –4.70 to 8.26; $I^2 = 74\%$; 3 studies, 397 participants; Analysis 1.11.1).

Figure 9.



Two studies reported individual symptoms (EORTC-55971; SCORPION). For clarity, in this update, we have presented data separately for QoL where higher values indicate better outcomes (functional scores; Analysis 1.9; Analysis 1.11), and those where higher scores indicate worse symptoms (symptoms scores; Analysis 1.10; Analysis 1.12). Overall, symptom scores may be worse at six months with PCRS (Analysis 1.10), and functional scores may be better with NACT (Analysis 1.9). By 12 months, EORTC QLQ-C30 scores were similar between groups for both functional scores (Analysis 1.11), and symptom scores (Analysis 1.12), probably reflecting more the effect of the disease than of treatment on patient-reported outcomes. However, due to the high dropout rate and heterogeneity, especially by 12 months, this evidence was of very low certainty and should be interpreted with caution. Only SCORPION reported EORTC QLQ-OV28 scores in an extractable form, and these were similar to the EORTC QLQ-C30 scores; they may favour NACT at 6 months, but are similar by 12 months. Please see SCORPION for details of these results.

Overall, the differences in quality-of-life outcomes were small and therefore unlikely to be clinically meaningful.

Other outcomes

Extent of residual disease

These data are reported based on those who underwent surgery and do not include those who did not.

In CHORUS, 79/219 women (36%) and 39/255 women (15%) had NMRD in the NACT and PCRS arms, respectively; 68/219 (31%) and 57/255 (22%) had SVRD in the NACT and PCRS arms, respectively; and 54/219 (25%) and 137/255 (54%) had LVRD (defined as > 1 cm) in the NACT and PCRS arms, respectively. Overall, 147/219 (67%)

women and 96/255 (38%) women in the NACT and PCRS arms, respectively, had either NMRD or SVRD.

In EORTC-55971, of those who had CRS, 151/295 women (51.2%) and 61/315 women (19.4%) had NMRD in the NACT and PCRS arms, respectively; 87/295 (29.5%) and 70/315 (22.2%) had SVRD in the NACT and PCRS arms, respectively. See [Supplementary material 2](#) table for further details. Therefore, of those who had NACT and ICRS, 238 women (80.7%) had either NMRD or SVRD compared to 131 women (41.6%) who had PCRS.

In Onda 2016, following initial surgery (PCRS or ICRS), 83/150 women (55%) and 45/147 women (31%) had NMRD in the NACT and PCRS arms, respectively; 24/150 (16%) and 47/147 (32%) had SVRD in the NACT and PCRS arms, respectively. Overall, following initial surgery (PCRS or ICRS), 107/150 women (71%) and 92/147 women (63%) had either NMRD or SVRD in the NACT and PCRS arms, respectively. Higher NMRD/SVRD rates than CHORUS and EORTC-55971 may be due to lower initial disease burden, since the entry criteria included all stage III disease, not just bulky stage IIIC.

In SCORPION, 57/74 women (77%) and 40/84 women (47.6%) had NMRD in the NACT and PCRS arms, respectively, and 16/74 (21.6%) and 38/84 (45.2%) had SVRD in the NACT and PCRS arms, respectively. Therefore, NMRD or SVRD was achieved for 73/74 (98.6%) and 78/84 (92.8%) in the NACT and PCRS arms, respectively. This is despite extensive pre-assessment and intraoperative exclusion (laparoscopic assessment) of those in whom NMRD was thought achievable, which differed significantly from CHORUS, Onda 2016, and EORTC-55971.

In TRUST, 271/320 (84.7%) and 235/334 (70.4%) had NMRD in the NACT and PCRS arms, respectively; 36/320 (11.3%) and 64/334 (19.2%) had SVRD in the NACT and PCRS arms, respectively. Overall,

307/320 (95.9%) and 299/334 (89.5%) had either NMRD or SVRD in the NACT and PCRS arms, respectively.

Chemotherapy-related adverse effects (G3+AEs) (Analysis 1.15)

Chemotherapy-specific-related toxicity was not specifically reported in EORTC-55971, as all severe adverse events (G3+) were reported together. However, the median time to restart chemotherapy after surgery was 18 days (range 5 to 55) and 19 days (range 0 to 84) in the NACT and PCRS groups, respectively. In SCORPION, the median time to start chemotherapy following surgery was lower in the NACT group (NACT = 27 days (range 16 to 37 days) versus PCRS = 40 days (range 17 to 120 days); $P < 0.001$) for the initial 110-patient cohort.

Two trials, assessing 768 women, found that there may be little or no difference in chemotherapy-related G3+ AEs between arms, although the evidence was of low certainty (OR 0.88, 95% CI 0.57 to 1.36; $I^2 = 54\%$; 2 studies, 768 participants; low-certainty evidence; [Supplementary material 6](#); Analysis 1.15; CHORUS; Onda 2016).

Duration of operation

Mean operating times in the SCORPION study for ICRS after NACT and PCRS were 253.2 minutes (SD = 101.4) and 460.6 minutes (SD = 102.6), respectively.

In EORTC-55971, the median operating times were 180 minutes (range 30 minutes to 560 minutes) and 165 minutes (range 10 minutes to 720 minutes) in the ICRS and PCRS arms, respectively. CHORUS reported that the median operation time was 120 minutes in both groups (interquartile ranges were 80 to 161 and 90 to 155 in the PCRS and NACT arms, respectively; the overall range was 12 to 450 minutes). Onda 2016 found that median operating time, when accounting for the first procedure only, was 302 minutes in the NACT group and 240 minutes in the PCRS group ($P < 0.001$). Due to disparities in the data collected, we were not able to combine these in a meta-analysis.

Median duration of surgery in the TRUST study for ICRS following NACT was 284 minutes (range 213 minutes to 360 minutes) and for PCRS was 331 minutes (range 253 minutes to 432 minutes).

Length of stay following surgery

SCORPION reported the mean length of hospital stay as 6.7 days (SD = 3.9 days) in the NACT group and 14.8 days (SD = 11.3) in the PCRS group ($P < 0.001$). In CHORUS, length of stay was provided as follows: "fewer women were discharged from hospital within 14 days after surgery in the primary-surgery group compared with primary chemotherapy (198/249, 80% versus 197/211, 93%, $P < 0.0001$)". Due to disparities in the data collected, data were not amenable to meta-analysis. These data were not available for EORTC-55971, Onda 2016, or TRUST.

Equity assessment

All studies enrolled participants limited to the female sex. Studies included participants from Algeria, Argentina, Austria, Belgium, Canada, Denmark, Egypt, France, Germany, Ireland, Italy, Japan, the Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, UK, and the USA. We did not investigate equity-related characteristics in this review.

Reporting biases

In SCORPION, quality-of-life data were not reported for all participants, with a greater proportion returning 12-month data in the NACT arm. Short-term postoperative severe adverse events (G3+) are reported more fully for the initial 110-participant cohort than in later versions with more participants. We have therefore kept reporting bias as unclear.

In CHORUS, the risk of reporting bias was unclear. All prespecified outcome measures were reported in some capacity, but quality-of-life data were provided only in the form of a global score at baseline and 6 and 12 months post-treatment.

We assessed Onda 2016 at low risk of reporting bias; surgical morbidities were reported in the initial publication, and survival outcomes were published in a subsequent update.

We assessed EORTC-55971 at unclear risk of reporting bias regarding quality-of-life data. EORTC-55971 (including Greimel and colleagues) subsequently published the quality-of-life data from the EORTC-55971 study [88, 89]. They reported that compliance for all women was too restrictive, and made changes to the protocol-defined analysis plan. The dataset for quality-of-life data was then restricted to institutions with the best compliance. The study authors stated that the sample size of the EORTC-55971 was overpowered to detect clinically meaningful differences in quality of life between study arms, therefore they decreased the sample size for quality-of-life data to 400 participants. They further restricted quality-of-life data collection to institutions that had 50% compliance at baseline and at least 35% on further follow-up over all enrolled women. Those participants in institutions selected for inclusion in quality-of-life data analysis had a greater median overall survival (9 months longer; $P = 0.001$) and a greater median progression-free survival (2.4 months longer; $P < 0.001$) than participants in the institutions that were not included in the quality-of-life data collection. Consequently, there was unclear risk of reporting bias regarding quality-of-life data, given the differences in disease that those participants selected for measurement of this outcome had in comparison with participants in the institutions not selected.

We assessed TRUST at unclear risk of reporting bias, as only the main outcomes of the study have been published in abstracts with further, incomplete unpublished data provided by the study authors from their meeting presentation.

DISCUSSION

Summary of main results

Seven studies involving a total of 2650 randomised participants met the inclusion criteria. Two studies were at critical risk of bias, and we excluded data from these studies from the meta-analyses. Following post-randomisation exclusions, the remaining five studies included 2380 participants in intention-to-treat analyses. See [Summary of findings 1](#) for further details.

We found little or no difference in overall survival outcomes in women with stage III/IV ovarian cancer who were treated with NACT plus ICRS compared with PCRS plus adjuvant chemotherapy. The certainty of evidence for this outcome is high, as it is unlikely to be affected by lack of blinding. The results for progression-free survival were similar, with little to no difference between groups,

although the certainty of evidence was moderate due to studies being unblinded (participants and doctors knew to which group participants were allocated), so there was potential for bias in this (and other) outcomes. An unplanned sensitivity analysis suggests there may be a small benefit in progression-free survival in the PCRS arms in the more recent studies where participants were more highly selected and higher rates of NMRD were achieved. However, this did not translate to an improvement in overall survival and hazards were non-proportionate, with those in the PCRS arms having higher rates of death in the first 12–18 months.

NACT reduced the risk of postoperative death, based on high-certainty evidence, as blinding was unlikely to have had a significant effect on this outcome. Surgically related adverse effects (G3+) were higher in the PCRS group (e.g. haemorrhagic, infective, and thromboembolic adverse events). NACT prior to surgery probably has a large effect on reducing the need for stoma formation (by almost two-thirds) and probably reduces the need for bowel resection by half, based on moderate-certainty evidence.

Quality-of-life outcomes were incompletely reported and at high risk of bias due to lack of blinding. Quality of life may be better overall for those in the NACT group at six months, although the results were heterogeneous (possibly due to variability in the disease burden between studies and changes in surgical radicality), and any differences were small and unlikely to represent a clinically meaningful difference for patients; any differences had disappeared by 12 months. The certainty of the evidence for quality-of-life outcomes was very low due to the potential for bias in reporting and poor quality of reporting in some studies.

Limitations of the evidence included in the review

We consider the current evidence for our critical outcomes of overall survival and progression-free survival to be of high and moderate certainty, respectively. We assessed the evidence for postoperative mortality as high certainty, and for surgical morbidity and adverse events as overall or moderate certainty; we downgraded outcomes due to potential risk of bias from lack of blinding. The evidence for quality-of-life outcomes was of very low certainty, downgraded mainly due to inconsistency, imprecision, attrition, and incomplete reporting.

In an earlier version of this review, the evidence for the non-inferiority of NACT versus PCRS for advanced ovarian cancer may not have been as widely applicable to current clinical management, as only participants with stage IIIC/IV ovarian tumours (extra-pelvic disease larger than 2 cm) were included in EORTC-55971, and the majority of participants had extensive disease (metastatic lesions larger than 10 cm were present in 61.6% of women) [49]. In the subgroup of women with preoperative extra-pelvic tumour of less than 5 cm in diameter (189 women), PCRS significantly improved overall survival compared with NACT (EORTC-55971). Furthermore, when subgrouped by FIGO stage, women with stage IV disease may have a survival advantage with NACT than with PCRS, although these data should be interpreted with caution due to inconsistencies between studies.

This update includes data from five studies with differing inclusion criteria, disease volume, and surgical approach, so the evidence for non-inferiority of NACT-ICRS compared with PCRS and adjuvant chemotherapy is more widely applicable.

Meta-analysis of five studies, assessing 2380 women, showed little to no difference in overall survival between NACT and PCRS for initial treatment for advanced ovarian cancer based on high-certainty evidence (CHORUS; EORTC-55971; Onda 2016; SCORPION, TRUST). The heterogeneity in overall survival and progression-free survival outcomes between studies were very low, despite criticisms about lack of surgical radicality in some of the older studies, and the effect did not change after exclusion of studies with lower surgical radicality on sensitivity analysis.

Meta-analysis of five trials showed there is likely little or no difference in risk of disease progression between NACT and PCRS for initial treatment for advanced ovarian cancer based on moderate-certainty evidence. However, in more highly selected patients (those with disease thought to be macroscopically resectable) in more recent studies where higher rates of NMRD were achieved with increased surgical radicality, there may be a slight improvement in progression-free survival with PCRS. However, these data should be interpreted with caution due to inconsistencies between studies and unplanned sensitivity analyses.

The quality of life data analysis of variance, adjusted for baseline scores, showed that there may be a difference in functional and symptom scores between groups at six months. However, the evidence is of very low certainty, and it is unlikely that the difference is clinically meaningful. The evidence for this outcome at 12 months is of very low certainty, due to high numbers of women dropping out, although the overall differences between the groups have been lost, suggesting the earlier differences may have been due to effects of the treatment. There may be little to no difference in global quality-of-life scores at six months, and by 12 months these results are very uncertain due to high levels of attrition.

Heterogeneity of disease burden and treatments between studies

These studies span a considerable time period (studies started recruiting from 1998 onwards). Surgical techniques and ethos, as well as chemotherapy regimens and maintenance treatments, have changed over the past three decades, with increased radicality of approach, as has the definition of so-called 'optimal' surgery (from < 2 cm deposits to NMRD now). Advances in cytoreductive surgery have enabled more extensive procedures to achieve higher rates of NMRD, but with a resulting increase in postoperative morbidity.

However, results of the more recently conducted studies, with higher surgical complexity, were consistent with those in older studies.

Disease burden

One of the criticisms of the EORTC-55971 and CHORUS studies has been that the NMRD rates for both arms were lower than those reported in retrospective cohort studies. However, EORTC-55971 and CHORUS both included women with extensive disease: ~70% of women in each arm with metastatic deposits measuring more than 5 cm, and a quarter of all participants had stage IV disease (EORTC-55971 specifically excluded stage IIIC disease based on para-aortic or pelvic lymph node metastases alone unless para-aortic lymph nodes were larger than 2 cm). In EORTC-55971, 61% of women in the PCRS arm had individual metastatic deposits larger than 10 cm (74% larger than 5 cm). Ten women in the PCRS arm and 19 women in the NACT/ICRS arm in CHORUS were unable to receive either study treatment due to disease burden. This is similar

to Onda 2016, where almost a third of women had stage IV disease. This is likely to represent the surgical equipoise at that time, so women with more extensive disease, thought to be less likely to have NMRD after surgery, were entered into the studies, and women with disease thought amenable to surgery were not enrolled. This contrasts with SCORPION, where fewer women had stage IV disease (13 (15.5%) in the PCRS arm versus 8 (9.2%) in the NACT/ICRS arm). Furthermore, women were only included in SCORPION if it was thought they could achieve NMRD or SVRD with CRS at laparoscopy. The TRUST study only included those with stage IIIC and IV disease thought to be macroscopically resectable to NMRD.

The SCORPION and TRUST studies are therefore not representative of many women with advanced ovarian cancer, who may be older or frailer, or both, which limits its applicability when examined in isolation. However, the shift towards including younger, fitter participants, with more favourable disease burden, may reflect a general shift in the point of equipoise, between NACT and PCRS, following results of the earlier studies, allowing those who believed in the superiority of PCRS to take part in later studies, and excluding those participants in whom there was no longer equipoise (with disease thought not to be resectable to NMRD).

Additional cytoreductive surgery

In the Japanese multicentre Onda 2016 study, in which 147 women underwent PCRS, NMRD or SVRD was achieved in 37% of women. More than a third of women in the PCRS arm underwent an additional attempt at CRS (additional CRS (ACRS)), despite maximal surgical effort at initial surgery, taking the total NMRD/SVRD proportion (< 1 cm residual disease) to 63% in the PCRS arm (PCRS + ACRS after four cycles of chemotherapy). This is a significant amount of additional treatment in the PCRS arm compared to the NACT/ICRS arm and puts the study at high risk of performance bias, since these women received additional treatment compared to those in the NACT arm, which was selectively delivered, as the study participants and personnel were not blinded. A proportion of women in the Onda 2016 and EORTC-55971 studies underwent PCRS and ACRS (37% and 17%, respectively) (after four cycles of chemotherapy in Onda 2016 and six cycles in EORTC-55971).

Residual disease volume

The SCORPION trial was a mono-centric trial that only randomised women to the trial once they had undergone a staging laparoscopy that produced a PI score of disease burden between ≥ 8 and ≤ 12 , predictive of achieving NMRD [128]. Not surprisingly, the NMRD rates achieved in the SCORPION study were higher than those of earlier studies: 90.9% of women in the PCRS arm achieved NMRD or SVRD (45.5% NMRD) compared with 90.4% in the NACT-ICRS arm (57.7% NMRD). The improved median overall survival of up to 43 months in SCORPION, in comparison with 27 months from the individual patient meta-analysis of the EORTC and CHORUS trial [81], likely represents differences in age, disease burden, and additional chemotherapy agents (notably bevacizumab and Poly (ADP-ribose) polymerase (PARP) inhibitors), as well as a shift in surgical ethos since the first study commenced. This is pertinent, as the EORTC-55971 study found in further analyses that NACT particularly benefited women with stage IV disease with individual metastatic deposits of 4.5 cm or larger, whereas PCRS may be preferable for those with stage IIIC disease and individual metastatic deposits smaller than 4.5 cm [129]. The more general applicability of the SCORPION trial is therefore compromised

by selecting only those who are deemed to have the potential for NMRD/SVRD rather than all-comers. TRUST was also highly selective, with likely achievement of NMRD one of the inclusion criteria. Despite this, and a surgical quality assurance process for participating centres, only 79% in the NACT arm and 68% in the PCRS arm had NMRD after surgery.

Additionally, although cytoreduction to NMRD is associated with a survival advantage, given that, to date, there has been no RCT comparing PCRS or NACT followed by ICRS to chemotherapy alone, by not attempting any surgical treatment on the subset of women who had very bulky disease, it is unclear if any differences in overall survival or progression-free survival would have been apparent, if they had been included in the trial. It is therefore interesting that, despite differences in patient selection and subsequent treatment, findings were largely similar across studies. This adds to the overall applicability of the findings of this meta-analysis.

Limitations due to applicability to clinical practice

Performance status and age

Importantly, studies recruited women who were, on average, younger and fitter (better ECOG performance status) than the general female population with EOC. This should be considered when making decisions at an individual level, since not all people with EOC would be eligible for inclusion in these studies.

Histological subtype

High-grade serous cancer of the ovary (HGSC) is the most common subtype of EOC and the most chemosensitive. The majority of participants had HGSC (CHORUS (70.5%); EORTC-55971 (61.8%); Onda 2016 (72.1%); SCORPION (97.7%); TRUST (92%)), and few had other, rare EOC subtypes. Very few women with low-grade serous cancers were enrolled (specifically excluded from TRUST). Although EORTC-55971 performed subgroup analysis for overall survival by histological subtype and found no significant differences, due to the small number of participants with other histological subtypes, we cannot be certain whether these results apply to those without HGSC, and people with less chemo-sensitive EOC may benefit more from PCRS.

Limitations due to lack of anti-angiogenic treatment and Poly(ADP-ribose) polymerase (PARP) inhibitor treatment and maintenance

Older studies did not include participants who were treated with anti-angiogenic agents (e.g. bevacizumab) [130], or PARP-inhibitors [131], although about a third of participants in SCORPION, and nearly two-thirds in TRUST were given bevacizumab during first-line treatment (59% in the NACT and 65% in the PCRS arm in TRUST). In TRUST, 30% in the NACT arm and 25% in the PCRS arm received a PARP-inhibitor.

Limitations due to lack of racial and ethnic inclusivity in studies

We did not address equity implications for practice in EOC. The included studies covered populations in Algeria, Argentina, Austria, Belgium, Canada, Denmark, Egypt, France, Germany, Ireland, Italy, Japan, the Netherlands, New Zealand, Norway, Portugal, Spain, Sweden, the UK, and the USA. However, the vast majority of participants were white, and those from other ethnicities are underrepresented in the data.

Limitations of the review processes

We have attempted to minimise biases in the review process. There is a potential for bias due to the introduction of subgroup and sensitivity analyses (i.e. stage, age, and residual disease) in a previous update of the review that were not specified in the original protocol. However, the decision to introduce subgroup analyses was made prior to the inclusion of the majority of studies included in the review. The decision for sensitivity analyses, based on NMRD rates, was made during this update process, but is based on discussion of differences in more recent studies in a previous version of the review.

We have been conducting research in this area for over 20 years and have run multiple searches over this time period. The risk of missing a study is therefore minimised.

An author of previous versions of this review [45, 46, 47, 48, 49], who was involved in a study included in this and previous updates (CHORUS), had no role in screening, decisions about inclusion or exclusion, data extraction, or analysis. This author was not involved in this or the most recent update [50]. Authors of the included studies have been helpful in providing data not included in the published version of their studies to minimise the risk of bias from missing results.

Data from TRUST are limited to those provided by the study authors in presentation and abstract form. It is likely that further data will be available on publication of the results. We still hope to include data from the Kumar 2009 trial. However, at the time of writing, the investigators had not published their final analysis, despite the trial being scheduled to be completed by 2012. We made the decision to discuss the interim data from this trial in [Agreements and disagreements with other studies or reviews](#) rather than as an included trial with incomplete outcomes to avoid the potential for bias in the results. Once these data are published, along with the results of the other ongoing trials (NCT04257786; NCT04515602; SUNNY) and peer-reviewed publication of TRUST, we plan to update the review.

Agreements and disagreements with other studies or reviews

Other studies

Investigators of the ongoing study, Kumar 2009, have presented interim results (at the American Society of Clinical Oncology (ASCO) conferences in 2006 and 2007) and the trial was scheduled for completion in 2012. Preliminary data from Kumar 2009 appear to corroborate the findings of the other studies included in this review. We understand from correspondence with Professor Kumar (from September 2011 to January 2012 and again in January 2019) that this trial is now closed. Owing to insufficient data in the 2009 report and discrepancies in some of the reported findings over time, we made the decision to await the final statistical analyses before including the interim data in meta-analyses (see [Supplementary material 5](#)).

Per-protocol pooled analysis of individual participant data from two included studies

One study pooled longer-term survival data from women in the CHORUS and EORTC-55971 studies, included as an additional reference to the studies [81]. This was a pre-planned analysis

prior to the launch of the CHORUS study. A total of 1220 women were included in the per-protocol pooled analysis (670 from EORTC-55971 and 550 from CHORUS), of whom 612 women received PCRS and 608 NACT. Median follow-up was 7.6 years. When women from both studies were combined, there was little or no difference in overall survival between groups (HR 0.97, 95% CI 0.86 to 1.09; $P = 0.586$). However, women with stage IV disease may have better overall survival and progression-free survival outcomes with NACT versus PCRS (overall survival HR 0.76, 95% CI 0.58 to 1.00; $P = 0.048$; progression-free survival HR 0.77, 95% CI 0.59 to 1.00; $P = 0.049$).

Systematic reviews

Systematic reviews of RCTs

A 2013 meta-analysis by Dai-yuan 2013 [132] examining the role of ICERS in ovarian cancer combined the RCTs of EORTC-55971 and that of Rose and colleagues [133]. However, the Rose study randomised women who had undergone PCRS and three cycles of chemotherapy to undergo a further ICERS prior to completing three further cycles of chemotherapy or to complete three further cycles of chemotherapy without further ICERS.

A systematic review by Yang 2017 [134] included four studies included in this review in their meta-analysis of serious adverse-event and quality-of-life data, but not survival data (CHORUS; EORTC-55971; Onda 2016; SCORPION). They showed that the NACT group had lower risks of grade 3/4 infections (RR 0.30, 95% CI 0.16 to 0.56), gastrointestinal fistulae (RR 0.24, 95% CI 0.06 to 0.95), risk of any grade 3 or 4 event (RR 0.29, 95% CI 0.11 to 0.78), and a lower rate of death within 28 days (RR 0.14, 95% CI 0.04 to 0.49), although with a similar risk of blood transfusion (RR 0.60, 95% CI 0.28 to 1.29). These findings are very similar to the current review.

Systematic reviews of RCTs and NRS

A systematic review and meta-analysis by Xiao 2018 [135] combined EORTC-55971 with nine cohort studies and two case-control studies. Given the inclusion of observational studies in this review, there is likely to be critical risk of selection bias in the NACT group, as the NACT group contained older women with more comorbidities, poorer performance status, higher CA125 at presentation, and later FIGO stage, compared to the PCRS group. The ORs produced for serious adverse events were in favour of NACT, although only major infection rates, wound complications, and vascular events reached statistical significance.

A meta-analysis by Qin 2018 [136] combined CHORUS and EORTC-55971 with 22 observational studies: 21 retrospective cohorts and one case-control study. The fixed-effect meta-analysis combining CHORUS and EORTC-55971 produced HRs for overall survival of 0.93 (95% CI 0.81 to 1.06) and for progression-free survival of 0.97 (95% CI 0.86 to 1.09), suggesting little or no difference between the two groups, similar to the findings of this review. The authors included data from a trial by Mellis and colleagues [137], but this study has subsequently been withdrawn from publication.

A meta-analysis by Zeng 2016 [138] combined four RCTs, but, like Dai-yuan 2013 included different treatment strategies in the NACT/ICERS arm: PCRS versus NACT/ICERS followed by completion of chemotherapy (CHORUS; EORTC-55971); PCRS followed by chemotherapy with randomisation to either further CRS (if

progressive disease ruled out) and completion chemotherapy or completion chemotherapy alone ([133] and Van der Burg 1995 [139]). This meta-analysis produced HRs for overall survival of 0.94 (95% CI 0.81 to 1.08) and for progression-free survival of 0.89 (95% CI 0.77 to 1.03).

A systematic review and meta-analysis focusing on factors predictive of postoperative morbidity included data from 15 articles, comprising 15,325 women with ovarian cancer, in the final meta-analysis [140]. Most of the studies were retrospective cohort studies. One study was a prospective cohort study. All studies were considered to be of "high quality." Two studies showed that NACT conferred a lower risk of developing postoperative complications (OR 0.44, 95% CI 0.28 to 0.70; $P < 0.001$; $I^2 = 0\%$), which is consistent with the direction of our findings.

Economic analyses

We did not specifically perform a search for articles examining the health economic effects of PCRS versus NACT. However, our search found five studies that compared the approaches in a variety of settings. We have therefore discussed their results as a brief economic commentary and will consider formal economic analysis in future updates of this review.

Cost-effectiveness analyses modelled from RCT data

Rowland 2015 [141] evaluated the cost implications of NACT versus PCRS, limiting their analysis to those over 65 years of age. The authors modelled their analyses on subgroup analyses based on age from EORTC-55971. They concluded that NACT produced cost savings of United States Dollars (USD) 5616 based on 2010 USA Medicare reimbursement rates at that time (calculated as equivalent to USD 7947/Euro (EUR) 7316/GBP pounds (GBP) 6258 in 2024).

A later cost-effectiveness study (Tran 2018 [142]), used data from four studies included in our meta-analysis (CHORUS; EORTC-55971; Onda 2016; SCORPION), to model costs of NACT versus PCRS, based on a hypothetical cohort of women aged 65 years with advanced EOC of median baseline characteristics for women in the USA. They estimated that NACT costs USD 20,762 per woman compared with USD 27,796 for PCRS, saving USD 7034 per woman in the seven-month post-treatment time horizon (calculated as equivalent to USD 9157/EUR 8431/GBP 7211 in 2024).

The same team modelled the costs of NACT and PCRS based on the more extensive surgical paradigm employed in SCORPION (Cole 2018 [143]). They found that NACT was associated with an estimated USD 142 million cost savings (calculated as equivalent to USD 184.9 million/EUR 170.2 million/GBP 145.6 million in 2024) based on the 15,000 women cohort. There were estimated to be 1098 fewer ovarian cancer-related deaths, 1355 additional life-years, and 1715 additional quality-adjusted life years. NACT was associated with a predicted cost saving of USD 9452 per woman (calculated as equivalent to USD 12,305/EUR 11,329/GBP 9690 in 2024) and a 7.3% lower risk of postoperative death.

Yang and colleagues conducted a systematic review comparing NACT/ICRS with PCRS and found four studies with 1607 participants [144]. Their outcomes were mainly short-term, including degree of cytoreduction and perioperative complications, but also included quality of life. They included four of the studies included in the meta-analysis of this review, and their findings were similar.

Another systematic review [145] included six RCTs with a total of 1901 participants, as they included both Chekman 2015 and the interim results reported in abstract only by Kumar 2009. Their results aligned with those of this study and concluded that there was high-certainty evidence of little difference in overall survival or progression-free survival between groups.

Other studies and reviews

Many review articles and non-randomised cohort studies have been published on this subject, many representing single-institution cohorts and including criticisms of the studies included in this review. Many of these studies are at critical risk of selection bias, especially as they do not examine all patients within a population, including those not fit for surgery, and so are likely to overestimate the benefits of PCRS [25]. This emphasises the importance of focusing on what is known from randomised data, where attempts have been made to limit these significant risks of bias. Please refer to the literature, since an in-depth narrative review of non-randomised studies is outside the scope of this review.

Based on the previously available data, there has been a shift to offering NACT in some treatment settings. A retrospective national cohort study reviewed US Surveillance, Epidemiology, and End Results (SEER) data from 1991 to 2007 for women with stage II-IV ovarian cancer [146]. Using regression analysis to adjust for effects of confounding variables on outcome and propensity score analysis to estimate the probability that a woman would undergo a given intervention, the authors performed a stratified analysis on women who lived longer than six months and underwent both surgery and chemotherapy in "high volume" centres. In the initial observational analysis, 5345 (55.8%) women underwent PCRS and 2238 (23.8%) underwent NACT; the remainder had no treatment.

The percentage of women undergoing NACT/ICRS increased from 19.7% in 1991 to 31.8% in 2007, with a concomitant decrease in PCRS from 63.2% in 1991 to 49.5% in 2007. Women most likely to receive NACT/ICRS were older, recently diagnosed (i.e. in the 2000s not 1990s), have serous histology, live in metropolitan areas, and have stage III or IV disease and a Charlson comorbidity score of 1. The substantial imbalance between treatment groups suggests strong selection bias in the cohort, and there were strong associations between area of residence in the USA and primary treatment received. An instrumental variable analysis was performed to assess for geographic variation in treatment pattern (the difference in the expected rates of NACT use and the observed rates of NACT use). Once this instrumental variable analysis was performed, the primary treatment chosen had minimal effect on cancer-specific survival (HR 0.94, 95% CI 0.58 to 1.52) or overall survival (HR 1.04, 95% CI 0.67 to 1.60). When the observational cohort and propensity-scored cohort survival data were calculated, these favoured PCRS (HR 1.27, 95% CI 1.19 to 1.35 and HR 1.24, 95% CI 1.15 to 1.34, respectively). The authors concluded that in the subset of women who had both surgery and chemotherapy (regardless of total cycles completed), there was no evidence of a difference in survival regardless of timing of surgery. The median overall survival in the propensity-scored cohort was 27.2 months in the PCRS group and 21 months in the NACT/ICRS group, not hugely dissimilar to EORTC-55971 data of 30 months in the NACT/ICRS group and 29 months in the PCRS group, emphasising the applicability of the RCT data included in the current review.

A Korean retrospective (2006 to 2014) cohort review of 435 consecutive women operated on in one centre looked at morbidity and survival differences after a paradigm shift in practice in 2010 to utilise more NACT/ICRS (Lee 2018 [147]). Comparing Group 2 (post-EORTC-55971) and Group 1 (post-EORTC-55971), HRs were 1.01 (95% CI 0.75 to 1.37) for progression-free survival and 0.93 (95% CI 0.63 to 1.36) for overall survival, with no differences in survival despite the increased use of NACT in Group 2. The authors concluded that use of NACT did not improve survival rates. However, there were no survival differences between groups after increased use of NACT, despite the women in Group 2 having more stage IV disease, more comorbidities, and more extensive surgery than women in Group 1.

A retrospective French study [148] from 15 gynaecological oncology specialist centres included 513 people with FIGO stage 3+ EOC, of which 334 could be matched by propensity scores based on pre-therapeutic variables. The authors found no difference in overall survival (HR 0.8, 95% CI 0.54 to 1.2) or median recurrence-free survival (PCRS 25.8 months (95% CI 22.6 to 29.9 months) versus ICRS 25.2 months (95% CI 20.6 to 35.8 months)). Interestingly, there was no difference in leaving NMRD after CRS (PCRS 85% versus ICRS 81.4%, $P = 0.76$), although rates of gastrointestinal tract resections (RR 1.61, $P < 0.001$), stoma (RR 2.38; $P = 0.003$), splenectomy (RR 2.5; $P = 0.02$), postoperative complications (RR 1.71; $P < 0.001$), and hospital stay were significantly higher in the PCRS group. These data, for patients operated on between 2000 and 2020, align with the findings of the current review. The PCRS group had almost double the rate of high-complexity surgery (29% versus 15%), with 50.8% bowel resection, 28% stoma formation, and 12% splenectomy rates in the PCRS group, suggesting a significant degree of 'maximal effort' surgery.

Melamed 2018 [149] conducted a quasi-experimental fuzzy regression discontinuity design and cross-sectional analysis comparing five regions in the USA. Two regions had rapidly increased their use of NACT in 2011 to 2012 by 27.3% and 23.3%, respectively. These regions were compared to three control regions where rates of NACT use in 2011 to 2012 had only increased by 2%. All-cause mortality in the increased NACT regions decreased (HR 0.81, 95% CI 0.71 to 0.94) compared to the control regions, which saw no change in all-cause mortality (HR 1.02, 95% CI 0.93 to 1.12). Death rates within 30 and 90 days of surgery also decreased in the regions that had increased NACT use (30-day mortality from 3.1% to 1.8% and 90-day mortality from 7.0% to 4.0%), which also differed from the control regions.

Melamed and colleagues went on to perform a difference-in-difference study in those providers in the USA who adopted a more liberal use of NACT post-EORTC-55971, compared with those who did not significantly change practice [44]. They identified 19,562 patients treated in 332 cancer centres that significantly increased their use of NACT (from 21.7% to 42.2%) and 19,737 patients who were treated in 332 centres that did not increase their use of NACT (20.1% to 22.5%). The standardised median overall survival improved by similar magnitudes in centres with high and low use of NACT after 2010 (difference-in-differences, 0.9 months, 95% CI -1.9 to 3.7). One-year mortality declined more in centres with high use of NACT than in those with low use (difference-in-differences, -2.1%, 95% CI -3.7 to -0.5). A more recent study looked at the effect on 90-day surgical mortality of utilisation of NACT by annual volume of CRS per centre (< 12 , 12 to < 24 , or ≥ 24 cases per year) [150].

As they expected from their previous work, higher NACT rates were associated with lower 90-day surgical mortality. What they did not expect was a greater decrease in high-volume centres with high NACT utilisation (OR 0.26; 95% CI, 0.17 to 0.41; 90-day mortality rates = 10.0% versus 2.9%) compared with average-volume centres (OR 0.49, 95% CI 0.33 to 0.72; 90-day mortality rates = 7.3% versus 3.7%) or low-volume centres (OR 0.48, 95% CI 0.39 to 0.60; 90-day mortality rates = 9.5% versus 4.8%). Improvements were not limited to short-term postoperative mortality, as in high-volume centres high NACT utilisation was associated with a 4.0 month improvement in five-year overall survival rates compared with low NACT utilisation (42.2 vs 38.1 months).

Conversely, a cohort study from Sweden found that altering their treatment paradigm to a maximal surgical effort, supra-regionalised surgical service, with a preference for PCRS, did not improve survival (HR 1.03, 95% CI 0.87 to 1.22; $P = 0.75$) [24]. Their change in practice reduced rates of NACT from 34% to 4% ($P < 0.001$). This was despite an increase in achieving NMRD (37% versus 67%; $P < 0.001$). Interestingly, the switch to a primary maximal CRS approach was also associated with an increase in non-surgically treated patients (24% versus 33%; $P \leq 0.001$). It would therefore appear that an 'all or nothing' approach to NMRD is not appropriate, and that patient selection to maximise those able to undergo combination treatment, be that with PCRS or NACT, is optimal at a population and individual level.

A comparative study from six centres in the UK found that age itself was not associated with poorer outcomes in older women with advanced EOC [151]. Although older patients were twice as likely to die (RR 1.98, 95% CI 1.63 to 2.39, $p < 0.001$), when adjusted for treatment received, women aged 75 or older did as well as younger women (relative risk 1.10, 95% CI 0.88 to 1.38, 99% reduction in χ^2 test). However, treatment (PCRS/adjuvant chemo, NACT/ICRS, surgery or chemotherapy alone or no treatment rates), varied significantly between sites, with some having no active treatment rates of 49% for patients aged 75 years or older (overall 'no active treatment' rates of 4% in < 75 versus 25% in those ≥ 75). The clinical skill therefore lies not just in surgical prowess, but in shared decision-making to optimise outcomes for each individual patient.

A study investigated patient preferences for attributes of PCRS versus NACT for the treatment of newly diagnosed ovarian cancer using a survey, educational video, and discrete choice experiment activities [152]. Overall, the 101 participants preferred better clinical outcomes, less extensive surgery, lower surgical mortality risks, lower risks of readmission, and longer progression-free survival and overall survival. Overall survival ranked the most important factor for consideration, followed by complications requiring readmission, progression-free survival, surgical mortality, extent of surgery, and lastly, treatment order. Participants would tolerate higher risks of operative morbidity and mortality to gain more substantial survival outcomes (six months). Conversely, participants were also willing to accept a reduction in survival outcomes (an 11-month reduction in progression-free survival (95% CI 5 to 19 months) and a 7-month reduction in overall survival (95% CI 2 to 12 months) to achieve a reduction in risk of surgical mortality.

We did not specifically analyse the effect of social, ethnic, and racial disparity in this review. However, several studies have. An audit of EOC care in England demonstrated significant differences in rates

of treatment for ovarian cancer between regions, including rates of surgery and combination of surgery and chemotherapy [41]. Similar results have been seen in other countries, with significant differences in access to high-quality surgery and survival outcomes associated with age, race, and ethnicity [153][154].

AUTHORS' CONCLUSIONS

Implications for practice

There is now a clearly established role for neoadjuvant chemotherapy before interval cytoreductive surgery versus primary cytoreductive surgery, now with five well-conducted studies, with differing inclusion criteria, demonstrating little or no difference in survival outcomes and reduced severe adverse events in those who received neoadjuvant chemotherapy. It is an area that has suffered from a distinct lack of equipoise, but the results of the SCORPION and TRUST studies cement these findings, and it is hoped that the debate can now move on to development of selection tools and shared decision-making to support women to make the best choices for themselves.

The current evidence is that a combination of chemotherapy and high-quality cytoreductive surgery with maximal tolerable effort is standard treatment for women with advanced ovarian cancer. The order of these treatment modalities appears to have little or no effect on survival outcomes for the overall population. These data support the role of primary cytoreductive surgery as treatment for advanced (stage IIIC/IV) ovarian cancer where achieving no macroscopic residual disease can be reasonably expected without causing excessive morbidity. Neoadjuvant chemotherapy should be the preferred alternative for women with stage IV disease, poor performance status, or comorbidities, due to the reduction in harms and their reduced ability to withstand major surgical procedures required to achieve no macroscopic residual disease. Compared to primary cytoreductive surgery, neoadjuvant chemotherapy may increase the rate of achieving no macroscopic residual disease, but this does not appear to translate into an increase in overall survival. In their discussion, the authors of SCORPION note that women with no macroscopic residual disease and those with small volume residual disease at primary cytoreductive surgery "have superimposable median progression-free survival". These data suggest that small-volume, chemotherapy-sensitive disease deposits are effectively treated by neoadjuvant chemotherapy, in line with another study [19].

The certainty of the evidence is high to moderate for overall and progression-free survival outcomes, respectively; high for perioperative death; moderate overall for adverse events; and low for quality-of-life outcomes. One important outcome for women to consider is that, based on these data, neoadjuvant chemotherapy reduces the risk of dying after surgery (2 fewer postoperative deaths for every 100 women having neoadjuvant chemotherapy compared to primary cytoreductive surgery) with high-certainty evidence. Neoadjuvant chemotherapy also likely reduces the risk by around two-thirds of needing a stoma following the operation, which may or may not be reversible later, depending on indication and subsequent response to treatment.

Cost-benefit analyses based on models derived from randomised controlled trial data suggest that a neoadjuvant chemotherapy strategy offers improved cost-effectiveness over a one-year time horizon following initial treatment, although these data will require

updating now that overall survival data are available from all the studies included in this review.

The clinical skill lies, therefore, not just in doing the best operation a patient can reasonably tolerate, but in joint decision-making to optimise outcomes for each individual. This is pertinent, especially as older and less fit participants are underrepresented in the studies, so the results may not be applicable to the general population of women with epithelial ovarian cancer.

Equity-related implications for practice

Access to specialist services and staff is required in order to reduce inequities in care for all women with ovarian cancer.

Implications for research

There are currently at least five ongoing studies (IRCT20250902067090N1 2025; NCT05371301; NCT04257786; NCT04515602; SUNNY), one unpublished randomised controlled trial (Kumar 2009), and four studies awaiting classification. We await the full publication of TRUST.

Collection of adverse-event and quality-of-life data are important patient-centred outcomes in advanced ovarian disease, especially if there is minimal difference in survival between treatment options. These were poorly or incompletely reported, or both, in studies included in this review. These data are increasingly important for shared decision-making, given the lack of difference in survival outcomes.

One study found that the quality of and participation within multidisciplinary team discussions varies significantly and is correlated with differences in first-line treatment decisions for people with ovarian cancer in the populations served [155]. One of the studies excluded from this review examined the role of a decision-making tool to help women reach an individual decision regarding intraperitoneal chemotherapy in ovarian cancer (Wenzel 2017 [156]). A similar tool, to aid shared decision-making with patients and their doctors, for treatment of women with advanced epithelial ovarian cancer would be extremely valuable and might increase the proportion of patients undergoing optimal treatment. Decision-aids should be developed to help deliver the optimal treatment for each individual, based on disease extent and performance status and comorbidities, regardless of age, race, ethnicity, or socio-economic status or educational attainment.

It is disappointing that there are as yet no final data published, or made available on direct request, for one of the randomised controlled trials (Kumar 2009). Data from studies that are long closed to follow-up, but remain unpublished in journals, should be made openly available on trials registries.

Most studies so far have offered interval cytoreductive surgery after three to four cycles of neoadjuvant chemotherapy. We do not yet know the optimal timing of interval cytoreductive surgery, although non-randomised studies suggest that delaying interval cytoreductive surgery with further cycles may compromise survival. However, these non-randomised studies are subject to critical risk of bias, since those who are less well initially would be more likely to have surgery after four to six cycles of chemotherapy. Future updates of this review should consider a network meta-analysis approach and include studies comparing different numbers of cycles of chemotherapy prior to interval

cytoreductive surgery (including so-called delayed debulking surgery) as well as primary cytoreductive surgery.

Equity-related implications for research

We did not specifically investigate equity-related characteristics in this review, although we did collect data on ethnicity and age. Despite ovarian cancer being most common in older women, few women over 70 years of age were included in any of the studies included in this review, and those with poorer performance status were underrepresented. This population is ill-served by clinical trials generally and, with an increasingly elderly population in many countries, this is an ever-expanding cohort of women for whom we have little evidence to support recommendations for treatment. Studies should also aim to recruit people regardless of race, ethnicity, or socio-economic status, as women of minoritised ethnicities are underrepresented in clinical studies.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD005343.pub8](https://doi.org/10.1002/14651858.CD005343.pub8).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of studies awaiting classification

Supplementary material 5 Characteristics of ongoing studies

Supplementary material 6 Analyses

Supplementary material 7 Data package

Supplementary material 8 Assessing 'Risk of bias' of included studies

ADDITIONAL INFORMATION

Acknowledgements

We thank the following people.

- Previous authors involved in the original protocol and review and review updates.
- Sean Kehoe: initial idea, supervisor and approval of the final version of the original review and previous review updates.
- Ignace Vergote and Corneel Coens for making available unpublished data from EORTC-55971 for the purposes of this review.
- Sean Kehoe and Matthew Nankivell for making available unpublished data from CHORUS.
- Christina Fotopoulou for making available unpublished data from TRUST.
- Professor Chekman for making available unpublished data from Chekman 2015 for the purposes of this review.
- Anna Fagotti for providing additional data for SCORPION.
- Takashi Onda for providing additional data for Onda 2016.
- JPH Tam for his invaluable translation and for assistance with communication with an author team in China.

- The Library Team (especially Natalie Parsley, Cate Newell, and Denise Manning (now retired)) of Somerset NHS Foundation Trust. Their continued support and endless enthusiasm for evidence-based medicine and quality improvement is invaluable, and we would like to offer our sincere thanks to them.
- Mojtaba Nouhi for obtaining the abstract list for the Iranian meeting found in the search strategy.
- We particularly thank the previous members of the Cochrane Gynaecological, Neuro-oncology and Orphan Cancer Group for years of work and dedication to producing high-quality evidence to support the use of evidence-based medicine in the gynaecological oncology patient and clinician community. We are all much diminished by your loss to Cochrane:
 - Jane Hayes for designing the original search strategy and Jo Platt for running top-up searches (and for her endless patience with us);
 - Tracey Harrison, Gail Quinn, and Clare Jess for their contribution to the editorial process.

This project was previously supported by the National Institute for Health Research (NIHR), via Cochrane Infrastructure funding to the Cochrane Gynaecological, Neuro-oncology and Orphan Cancer Group. The views and opinions expressed therein are those of the review authors and do not necessarily reflect those of the Systematic Reviews Programme, NIHR, NHS, or the Department of Health.

We are grateful to the following peer reviewers for their time and helpful comments on previous versions of this review: Lars Henning, Alexander Melamed, and Monique Spillman.

The authors sincerely thank Professor Paul Pharoah for his very helpful feedback on 29 January 2021 regarding an error in the October 2019 version of this review. None of us are perfect, and we are all (and science in particular) stronger for open, honest, and civil feedback.

Editorial and peer-reviewer contributions

The following people conducted the editorial process for this article.

- Sign-off Editor (final editorial decision): Toby Lasserson, Deputy Editor in Chief, Cochrane Central Executive
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Gail Quinn, Cochrane Central Editorial Service
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments, and supported the editorial team): Leticia Rodrigues, Cochrane Central Editorial Service
- Copy Editor (copy editing and production): Denise Mitchell, Cochrane Central Production Service
- Peer reviewers (provided comments and recommended an editorial decision): Katharina da Silva Lopes, Evidence Production and Methods Directorate (methods), Steve McDonald, Cochrane Australia (search).

Contributions of authors

- Catherine Elderfield: co-review author, screened search results, assessed papers for eligibility, extracted data, contributed to analysis, and co-wrote this updated review

- Mohamed Shawky: co-review author, screened search results, assessed papers for eligibility, extracted data and co-wrote the previous review update.
- Cherry Choudhary: co-review author, screened search results, assessed papers for eligibility.
- Sarah Coleridge: co-review author, screened original search results, assessed papers for eligibility, extracted data, and co-wrote previous review updates.
- Andrew Bryant: assisted with data extraction, data analysis, and writing of the final version of this and previous review updates.
- Ryan Beazley: screened search results, assessed papers for eligibility.
- Jo Morrison: co-review author, wrote protocol, screened search results, assessed papers for eligibility, extracted data, data-analysis, and co-wrote the original review and all its updates.

Declarations of interest

- Catherine Elderfield: no conflict of interest
- Mohamed Shawky: no conflict of interest
- Cherry Choudhary: no conflict of interest
- Sarah Coleridge: no conflict of interest (full-time NHS consultant with no private practice, practising both PCRS and NACT/ICRS)
- Andrew Bryant: no conflict of interest
- Jo Morrison: no conflict of interest; Senior Editor for Cochrane (Sexual and Reproductive Health Thematic group), although the author was not involved in the editorial process of this review; full-time NHS consultant with no private practice, practising both PCRS and NACT/ICRS

Sources of support

Internal sources

- New Source of support, UK

The review update was performed without formal internal support.

External sources

- 10/4001/12 NIHR Cochrane Programme Grant Scheme, UK

A previous update of the review received methodological and statistical support as part of the 10/4001/12 NIHR Cochrane

What's new

Programme Grant Scheme - Optimising care, diagnosis and treatment pathways to ensure cost effectiveness and best practice in gynaecological cancer: improving evidence for the NHS. All subsequent review updates have been performed without specific funding. No external support was received for this update.

Registration and protocol

Protocol (2005): DOI: 10.1002/14651858.CD005343

Original review (2007): DOI: 10.1002/14651858.CD005343.pub2

Review update (2012): DOI: 10.1002/14651858.CD005343.pub3

Review update (2019): DOI: 10.1002/14651858.CD005343.pub4

Review update (2021): DOI: 10.1002/14651858.CD005343.pub5

Review updated (2021a): DOI: 10.1002/14651858.CD005343.pub6

Review updated (2025): DOI: 10.1002/14651858.CD005343.pub7

Data, code and other materials

As part of the published Cochrane review, the following are made available for download for users of the Cochrane Library: full search strategies for each database; full citations of each unique report for all studies included, ongoing or awaiting classification, or excluded at the full-text screen, in the final review; study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; and analysis data, including overall estimates and settings, subgroup estimates, and individual data rows. Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, Review Manager Web, using the inbuilt computation methods. Template data extraction forms are available from the authors on reasonable request.

Disclosure of artificial intelligence use

We used the Cochrane RCT Classifier tool within Covidence to initially sift titles and abstracts to remove studies that did not appear to be clinical trials. We did not use AI for any other parts of this review.

Date	Event	Description
29 July 2026	New search has been performed	New search to 9 October 2025; new studies added
29 July 2026	New citation required but conclusions have not changed	Search updated - 2 new studies included.

History

Protocol first published: Issue 3, 2005

Review first published: Issue 4, 2007

Date	Event	Description
10 February 2025	New citation required but conclusions have not changed	Search updated - new data included for previously included study; no new studies included.
10 February 2025	New search has been performed	Search updated - new data included for previously included study; no new studies included.
7 April 2021	New search has been performed	New search to 9 October 2020, and data added from studies included in previous version
7 April 2021	New citation required but conclusions have not changed	Review updated but conclusions not changed.
1 February 2021	Amended	Correction to survival data for CHORUS
1 February 2021	New citation required but conclusions have not changed	New citation required, but conclusions have not changed. Correction to survival data for CHORUS
29 May 2019	New search has been performed	Search updated 11 February 2019.
28 May 2019	New citation required but conclusions have not changed	Updated with inclusion of four new studies. Three ongoing unpublished studies identified.
27 March 2014	Amended	Contact details updated.
21 June 2012	New search has been performed	Search updated; 26 newly identified reports added to studies awaiting classification, including five reports of three ongoing studies (CHORUS #a; Kumar #a; Onda #a).
21 June 2012	New citation required and conclusions have changed	One new trial (EORTC-55971) included. Conclusions changed.

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ADDITIONAL TABLES

Table 1. Carcinoma of the ovary: FIGO^a 1988 nomenclature

Stage	Extent of tumour	Substage	Details
I	Limited to ovaries	IA	Limited to 1 ovary, no tumour on surface or capsule rupture, no positive ascites
		IB	Limited to both ovaries, no tumour on surface or capsule rupture, no positive ascites
		IC	Stage Ia or Ib but with capsule ruptured, tumour on ovarian surface or positive peritoneal washings/ascites
II	Limited to 1 or both ovaries with pelvic extension	IIA	Extension, metastases to uterus, tubes, or a combination
		IIB	Extension to other pelvis tissues
		IIC	Stage IIa or Iib with tumour on the surface of 1 or both ovaries, or with capsule ruptured, or with positive peritoneal washings/ascites
III	Limited to abdomen with histologically confirmed peritoneal implants outside the pelvis or positive nodes, or both, or extension to small bowel or omentum	IIIA	Tumour grossly limited to the true pelvis with negative regional lymph nodes, microscopic seeding of abdominal peritoneal surfaces or extension to small bowel or mesentery
		IIIB	Macroscopic metastases < 2 cm; negative regional lymph nodes
		IIIC	Macroscopic metastases > 2 cm or positive regional lymph nodes, or both
IV	Distant metastases		Growth outside the abdominal cavity (e.g. lung, liver parenchyma (superficial liver metastases is stage III))

^aFIGO: International Federation of Gynecology and Obstetrics 1988 [157]; four included studies used pre-2014 classification (Chekman 2015; CHORUS; EORTC-55971; Onda 2016).

Table 2. FIGO^a staging 2014

Stage	Extent of tumour	Substage level 1	Details	Substage level 2	Details	Substage level 3	Details
I	Limited to ovaries	IA	Limited to 1 ovary, no tumour on surface or capsule rupture, no positive ascites				
		IB	Limited to both ovaries, no tumour on surface or capsule rupture, no positive ascites				
		IC	Stage Ia or Ib but with capsule ruptured, tumour on ovarian surface or positive peritoneal washings/ascites	IC1	Surgical spill		
				IC2	Capsule rupture before surgery or tumour on ovarian surface		
IC3	Malignant cells in the ascites or peritoneal washings						
II	Limited to 1 or both ovaries with pelvic extension	IIA	Extension and/or implant on uterus and/or fallopian tubes				
		IIB	Extension to other pelvic intraperitoneal tissues				
III	Limited to abdomen with histologically confirmed peritoneal implants outside the pelvis or positive nodes, or both, or extension to small bowel or omentum	IIIA	IIIA (positive retroperitoneal lymph nodes and/or microscopic metastasis beyond the pelvis)	IIIA1	Positive retroperitoneal lymph nodes only	IIIA1i	Metastasis ≤ 10 mm
						IIIA1ii	Metastasis > 10 mm
				IIIA2	Microscopic, extrapelvic (above the brim) peritoneal involvement ± positive retroperitoneal lymph nodes		
		IIB	Macroscopic, extrapelvic, peritoneal metastasis ≤ 2 cm ± positive retroperitoneal lymph nodes. Includes extension to capsule of liver/spleen				
		IIIC	Macroscopic, extrapelvic, peritoneal metastasis > 2 cm ± positive retroperitoneal lymph nodes. Includes extension to capsule of liver/spleen				
IV	Distant metastases	IV	Growth outside the abdominal cavity (e.g. lung, liver parenchyma (superficial liver metastases is stage III))	IVA	Pleural effusion with positive cytology		
				IVB	Hepatic and/or splenic parenchymal metastasis, metastasis to extra-abdominal organs (including inguinal lymph nodes and lymph nodes outside the abdominal cavity)		

^aFIGO: International Federation of Gynecology and Obstetrics revised staging 2014 [158], as used in SCORPION and TRUST. This was updated but unchanged in the 2025 update [159].

Table 3. Overview of included studies and syntheses

Study ID (country)	Study design	N randomised	FIGO stage NACT arm	FIGO stage PCRS arm	NACT Age (years)	PCRS Age (years)	No. of cycles NACT prior to ICRS	Interventions (NACT vs PCRS)	AE assessment and QoL scales	Notes
Trial number										
Abou-Taleb 2025 (Egypt, 1 centre)	Single-centre, randomised, unblinded	80	IIB = 1 (2.5%); III = 29 (72.5%); IV = 10 (25%)	IIB = 3 (7.5%); III = 33 (82.5%); IV = 4 (10%)	Mean = 57.23 (SD ± 9.61)	Mean = 55.37 (SD ± 9.67)	NR	NACT with bevacizumab + ICRS (arm IIA, experimental) and NACT without bevacizumab + ICRS (arm IIB, experimental): or PCRS + adjuvant chemotherapy with bevacizumab (arm IA) and PCRS + adjuvant chemotherapy without bevacizumab (arm IA)	NR	overall survival and progression-free survival at one year were recorded but primary outcomes were surgical resection and surgical complications. Survival curves only recorded following surgery
Chekman 2015 ^a (Algeria, 1 centre)	Single-centre, randomised, unblinded	82	100% stage IIIC	100% stage IIIC	NR	NR	NACT prior to ICRS: 3 cycles: 6 (14.6%); 4 cycles: 4 (9.6%) 6 cycles: 18 (43.9%); no surgery: 13 (31.7%)	NACT: 3 to 6 cycles chemotherapy before ICRS vs PCRS	NR	"90 women enrolled and underwent surgery". Limited data were provided in an abstract; further data provided by the author. See Chekman 2015 for details. The study is at high risk of bias due to study reporting, precluding inclusion of data. ^b
CHORUS (UK and New)	International, multicentre, open-	*550	Stage III = 206 (75%); Stage IV = 68 (25%)	Stage III = 206 (75%); Stage IV = 70 (25%)	Median = 65 (range 34 to 88)	Median 66 (range 26 to 87)	NACT prior to ICRS: surgery scheduled after 3 cycles of	NACT: 3 cycles chemotherapy + ICRS vs PCRS	NCI CTCAE version 3.0; EORTC QLQ-C30;	*by ITT ^b

Table 3. Overview of included studies and syntheses (Continued)

Zealand, 87 recruiting and 64 surgical centres)		label, non-inferiority phase III RCT					chemotherapy.		EORTC QLQ-OV28	
ISRCTN74802813							195/274 (71%) had surgery after 3 cycles			
EORTC-55971 International, multicentre, open-label, non-inferiority phase III RCT (Argentina, Belgium, Canada, Italy, the Netherlands, Norway, Spain, the UK, 59 centres)	670		Stage IIIC = 253 (75.7%); Stage IV = 81 (24.3%)	Stage IIIC = 257 (76.5%); Stage IV = 77 (22.9%)	Median 63 (range 33 to 81)	Median 62 (range 25 to 86)	NACT prior to ICRS: surgery scheduled after 3 cycles of chemotherapy. 295/334 (88%) had surgery (unclear after how many cycles)	NACT: 3 cycles chemotherapy + ICRS vs PCRS	NCI CTCAE version 3.0; EORTC QLQ-C30; EORTC QLQ-OV28	Only institutions with good QoL compliance were included in the QoL substudy. The institutions with good QoL compliance differed from those studies excluded from the QoL analysis. ^b The 48 Argentinian participants were excluded post-randomisation owing to authorisation irregularities at the Argentinian centre.
NCT00003636										
Onda 2016 (Japan, 34 centres) JCOG0602	301	Multicentre, open-label, non-inferiority, phase III RCT	Stage III = 105 (69%); Stage IV = 47 (31%)	Stage III = 100 (67%); Stage IV = 49 (33%)	Median 60.5 (range 36 to 75)	Median 59 (range 30 to 75)	NACT arm: 4 cycles pre-ICRS = 130/152; 4 cycles post-ICRS = 103/152 PCRS arm: 4 cycles after PCRS = 128/149; 4 further cycles = 99/152; *49/149 had additional ICRS after PCRS	NACT: 4 cycles chemotherapy + ICRS vs PCRS + further surgery for 49 participants	NCI CTCAE version 3.0	*49 in PCRS arm underwent a second attempt at ICRS after 4 cycles of adjuvant chemotherapy during primary treatment; residual disease status presented for first surgery only and by ITT. ^b
SCORPION	171	Single-institution, open-label	Stage IIIC = 71 (84.5%);	Stage IIIC = 79 (90.8%);	Mean = 56.1 (SD = 10.4)	Mean = 2368.4	NACT prior to ICRS: median	NACT: 3 to 4 cycles chemotherapy	NCI CTC version 2.0; EORTC	Surgical cytoreduction degree for initial 110 cohort only ^b

Table 3. Overview of included studies and syntheses (Continued)

(Italy, 1 centre)	bel, su- periority, phase III NCT01461850 RCT		Stage IV = 8 (9.2%)	Stage IV = 13 (15.5%)		(SD = 3199.3)	number of cy- cles 4 (range 3 to 4)	apy + beva- cizumab for selected par- ticipants vs PCRS	QLQ-C30; EORTC QLQ-OV28	
TRUST; Germany, Italy, the UK, France, USA, Swe- den, Den- mark, Aus- tria: 20 centres	Interna- tional, multicen- tre, open- label, non- inferiority phase III RCT	796 pa- tients (688 after post- randomi- sation ex- clusions) NACT-ICRS Ran- domised: n = 398 Excluded: n = 55, due to no ovarian cancer (n = 32); early stage (n = 21); general in- cl/excl cri- teria (n = 2). PCRS Ran- domised: n = 398 Excluded: n = 53, due to no ovarian cancer (n = 27); early stage (n = 16) general in- cl/excl cri-	IIIB = 18 (5.3%); IIIC = 217 (63%); IVA = 35 (10%); IVB = 68 (20%)	IIIB = 30 (8.7%); IIIC = 203 (59%); IVA = 31 (9.0%); IVB = 79 (23%)	Median age = 64 (range 32-83)	Median age = 63 (range 34-83)	Median age = 63 (range 34-83)	NACT prior to ICRS: surgery scheduled af- ter 3 cycles of chemothera- py.	Carbo- platin AUC5, pa- clitaxel 175 mg/ m2 3- weekly beva- cizumab 15 mg/kg 3-weekly as indicat- ed PARPi as indicated Study par- ticipation or any oth- er treat- ment as long as applic- able for both study arms	Measured by EORTC QLQ-C30 (Version 3), EORTC QLQ-OV28, EQ-5D-3L

Table 3. Overview of included studies and syntheses (Continued)

teria (n = 8) Death before interventions (n = 2)	CA125: carcinoma antigen 125; AE: adverse event; CI: confidence interval; CTC: Common Toxicity Criteria; CTCAE: Common Terminology Criteria for Adverse Events; ECOG: Eastern Cooperative Oncology Group; EORTC: European Organisation for Research and Treatment of Cancer; FIGO: International Federation of Gynecology and Obstetrics; ICRS: interval cytoreductive surgery; ITT: intention-to-treat; NACT: neoadjuvant chemotherapy; NCI: National Cancer Institute; NR: not recorded; OS: overall survival; PCRS: primary cytoreductive surgery; PFS: progression-free survival; QLQ-C30: quality of life questionnaire core-36; QLQ-OV28: ovarian cancer-specific quality of life questionnaire; QoL: quality of life; RCT: randomised controlled trial; SD: standard deviation
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^aWe are very uncertain of the data in Chekman 2015, which appear to be overall survival data, although they were presented as progression-free survival data (with other figures for progression-free survival of 13.15 months (95% CI 9.19 to 17.10) but do not include an early death after cycle 2 NACT). We have not included data from this study in the analyses due to concerns about data quality and risk of bias.

^bProgression-free survival and overall survival were reported in the study but are excluded from this table.